

Lectures on Monte Carlo Theory

Chapter 5: Variance Reduction Techniques (Part 1)

Based on Lorek & Rolski

Chapter 5 Outline (Part 1)

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Why Variance Reduction? The Fundamental Problem I

The simplest Monte Carlo estimator for the quantity $I = \mathbb{E}[Y]$ (the expected value, i.e., the long-run average, of a random variable Y) is called the *Crude Monte Carlo* (CMC) estimator. It averages R (capital R) independent copies of Y :

The Crude Monte Carlo Estimator

$$\hat{Y}_R^{\text{CMC}} = \frac{1}{R} \sum_{j=1}^R Y_j,$$

where $Y_1, Y_2, \dots, Y_R$ (Y sub 1, Y sub 2, ..., Y sub R) are independent, identically distributed (i.i.d.) replications of Y , R is the total number of replications (simulation runs), and $\hat{Y}_R^{\text{CMC}}$ (Y -hat sub R , CMC) denotes the CMC estimate of I .

The variance of this estimator is $\text{Var}(\hat{Y}_R^{\text{CMC}}) = \text{Var}(Y)/R$, which means the spread of our estimates around the true value I shrinks as we run more simulations — but only slowly, because it depends on $1/R$ rather than $1/R^2$ or faster.

Why Variance Reduction? The Fundamental Problem II

The Fundamental Problem: Diminishing Returns

The half-width of a 95% confidence interval for I is approximately

$$b \approx 1.96 \cdot \frac{\sigma_Y}{\sqrt{R}},$$

where σ_Y (sigma sub Y, pronounced “sigma sub Y”) is the standard deviation of Y — a measure of how much individual observations of Y scatter around the mean I .

This formula tells us that to halve the error b , we must quadruple R . In other words, going from $R = 1,000$ to $R = 4,000$ only cuts the error in half. To reduce the error by a factor of 10, we need 100 times as many runs.

The alternative: if we can construct a new estimator with variance $\text{Var}(Y)/k$ for some factor $k > 1$, it achieves the same precision as running k times more simulations — for free. That is the goal of variance reduction.

Why Variance Reduction? The Fundamental Problem III

Formally, any variance reduction method constructs a new estimator $\hat{Y}_R^{\text{new}}$ of I that must satisfy two requirements:

- 1 **Consistency:** $\hat{Y}_R^{\text{new}} \rightarrow I$ with probability 1 as $R \rightarrow \infty$. The estimator must converge to the correct answer given enough data.
- 2 **A Central Limit Theorem (CLT):**

$$\sqrt{R}(\hat{Y}_R^{\text{new}} - I) \xrightarrow{\mathcal{D}} \mathcal{N}(0, (\varsigma^{\text{new}})^2),$$

where $\sqrt{R}$ (square root of R) rescales the error to a non-trivial limit, $\xrightarrow{\mathcal{D}}$ (arrow with $\mathcal{D}$) means “converges in distribution,” $\mathcal{N}(0, (\varsigma^{\text{new}})^2)$ is a normal (bell-curve) distribution centred at 0 with variance $(\varsigma^{\text{new}})^2$, and ς (varsigma, a variant of the Greek letter sigma, pronounced “var-sigma”) is the *asymptotic standard deviation*.

The quantity $(\varsigma^{\text{new}})^2$ is called the *time-average variance constant* (TAVC) or *asymptotic variance*. It is defined as $(\varsigma^{\text{new}})^2 = \lim_{R \rightarrow \infty} R \cdot \text{Var}(\hat{Y}_R^{\text{new}})$. A smaller TAVC means that for the same number of runs R , we get a narrower confidence interval and a more precise estimate of I .

Why Variance Reduction? The Fundamental Problem IV

Eight Methods Covered in This Chapter

- ➊ **Stratified sampling** — divide the input space into strata and sample each separately.
- ➋ **Antithetic variates** — generate negatively correlated pairs that cancel out overestimates and underestimates.
- ➌ **Common random numbers (CRN)** — use the same random stream when comparing two systems, to reduce the variance of the difference.
- ➍ **Ratios of expectations (RoE)** — handle integrals that naturally appear as a ratio.
- ➎ **Control variates (CV)** — use a correlated auxiliary variable with a known mean to correct the estimate.
- ➏ **Conditional Monte Carlo** — compute a conditional expectation analytically, replacing noisy samples with smooth values.
- ➐ **Importance sampling (IS)** — sample from a different distribution that concentrates effort in the most important region.
- ➑ **Cross-entropy method (CE)** — systematically find the best importance sampling distribution from a parametric family.

Each method is presented with its theoretical guarantee, a worked numerical example, and Python code. 4/109

5.1.1 Stratified Sampling — Idea and Setup I

What problem does stratified sampling solve? When we draw R samples from the distribution of Y at random, most samples may land in easy, low-variability regions, while a few hard, high-variability regions are underrepresented. The crude Monte Carlo estimator can therefore be highly variable. The remedy is simple: instead of sampling from the full distribution of Y at once, we divide the sample space into non-overlapping *strata* (regions) and sample *separately* from each one. By pre-committing a fixed number of samples to each stratum, we prevent the accidental under-sampling of any region.

5.1.1 Stratified Sampling — Idea and Setup II

Setup (Definition 5.1)

Partition the sample space Ω (omega, the set of all possible outcomes) into m disjoint strata $S^1, S^2, \dots, S^m$ such that every outcome belongs to exactly one stratum: $\bigcup_{j=1}^m S^j = \Omega$ (the union of all strata is the whole space). Each stratum S^j has probability $p_j = \mathbb{P}(Y \in S^j) > 0$ (p sub j , the proportion of the population lying in stratum j), and together they sum to one: $\sum_{j=1}^m p_j = 1$. We define two key quantities for each stratum j :

- $I^j = \mathbb{E}[Y \mid Y \in S^j]$ (I superscript j): the conditional expectation of Y within stratum j , i.e., the average value of Y if we know Y falls in that stratum. This is generally unknown.
- $\sigma_j^2 = \text{Var}(Y \mid Y \in S^j)$ (sigma squared sub j): the within-stratum variance, measuring how much Y varies *inside* stratum j .

By the law of total expectation, the overall mean satisfies: $I = \mathbb{E}[Y] = \sum_{j=1}^m p_j I^j$ (the overall mean is a weighted average of the stratum means, with weights equal to the stratum probabilities).

5.1.1 Stratified Sampling — Idea and Setup III

Law of Total Variance (Proposition 5.1.1)

For any two random variables Y and J (where J randomly selects the stratum index, with $\mathbb{P}(J = j) = p_j$), the variance of Y decomposes as:

$$\text{Var}(Y) = \underbrace{\mathbb{E}[\text{Var}(Y \mid J)]}_{\text{average within-stratum variance}} + \underbrace{\text{Var}(\mathbb{E}[Y \mid J])}_{\text{between-stratum variance}}.$$

Expanding this formula in terms of σ_j^2 and I^j (Corollary 5.1.2):

$$\text{Var}(Y) = \underbrace{\sum_{j=1}^m p_j \sigma_j^2}_{\text{within-stratum}} + \underbrace{\sum_{j=1}^m p_j (I^j - I)^2}_{\text{between-stratum}}.$$

Here $\sum_{j=1}^m p_j \sigma_j^2$ (Sigma, meaning “sum over j from 1 to m ” of $p_j \sigma_j^2$) is the probability-weighted average of within-stratum variances, and $\sum_{j=1}^m p_j (I^j - I)^2$ measures how much the stratum means I^j differ from the overall mean I .

5.1.1 Stratified Sampling — Worked Example: Estimating π I

5.1.1 Stratified Sampling — Worked Example: Estimating π II

Example 5.1.8 — Estimating π via Stratified Sampling

We estimate π (pi, approximately 3.14159) using the area of a quarter-circle. If $U \sim \mathcal{U}[0, 1]$ (U uniformly distributed on $[0, 1]$), then:

$$Y = 4\sqrt{1 - U^2}, \quad \mathbb{E}[Y] = 4 \int_0^1 \sqrt{1 - u^2} du = \pi.$$

Here $\sqrt{1 - u^2}$ is the height of a unit circle at horizontal position u , so $4 \int_0^1 \sqrt{1 - u^2} du$ equals 4 times the area of a quarter-circle of radius 1, which is π .

We define m equally probable strata as intervals of length $1/m$: $B^j = (\frac{j-1}{m}, \frac{j}{m}]$ for $j = 1, \dots, m$, so $p_j = 1/m$ (each stratum has equal probability $1/m$).

To sample uniformly within stratum j , set $V^j = (j-1)/m + U/m$ where $U \sim \mathcal{U}[0, 1]$, and compute $Y^j = 4\sqrt{1 - (V^j)^2}$. The stratified estimator with proportional allocation ($R_j = R/m$) is:

$$\hat{Y}_R^{\text{str}} = \frac{1}{m} \sum_{j=1}^m \hat{Y}_{R_j}^j = \frac{1}{R} \sum_{j=1}^m \sum_{i=1}^{R/m} 4\sqrt{1 - (V_i^j)^2}.$$

Numerical results from Table 5.2 of the textbook, with total $R = 10^4$ (ten thousand) replications: 11 / 109

5.1.1 Stratified Sampling — Python Code

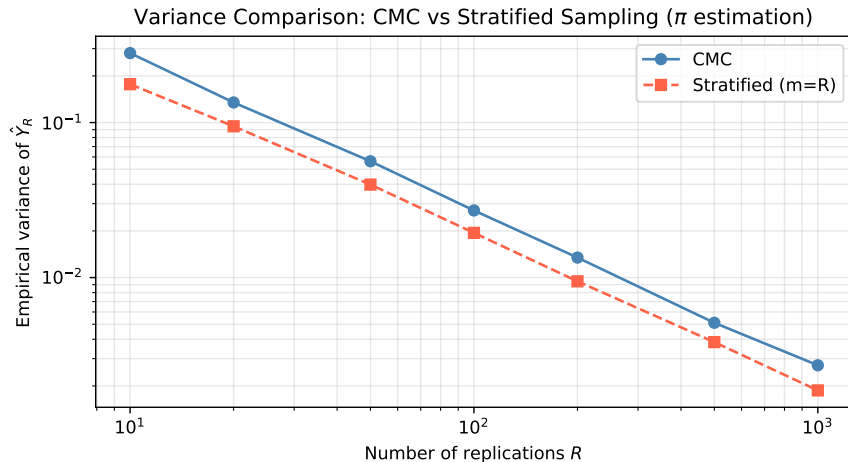
```
import numpy as np

def stratified_pi(m, R, use_optimal=False, R_pilot=100):
    """Estimate pi using stratified sampling on U with m strata."""
    if use_optimal:
        # Pilot phase: estimate sigma_j in each stratum
        R_j_pilot = R_pilot // m
        sigma_hat = []
        for j in range(m):
            V = (j + np.random.uniform(0, 1, R_j_pilot)) / m
            Yj = 4 * np.sqrt(1 - V**2)
            sigma_hat.append(np.std(Yj, ddof=1))
        sigma_hat = np.array(sigma_hat)
        # Optimal allocation: R_j proportional to sigma_hat[j]
        R_j = np.maximum(1, np.round(sigma_hat / sigma_hat.sum() * R).astype(int))
    else:
        R_j = np.full(m, R // m) # proportional (equal p_j = 1/m)

    estimates = []
    for j in range(m):
        V = (j + np.random.uniform(0, 1, R_j[j])) / m
        Yj = 4 * np.sqrt(1 - V**2)
        estimates.append(np.mean(Yj))
    return np.mean(estimates) # equal p_j = 1/m, so unweighted mean

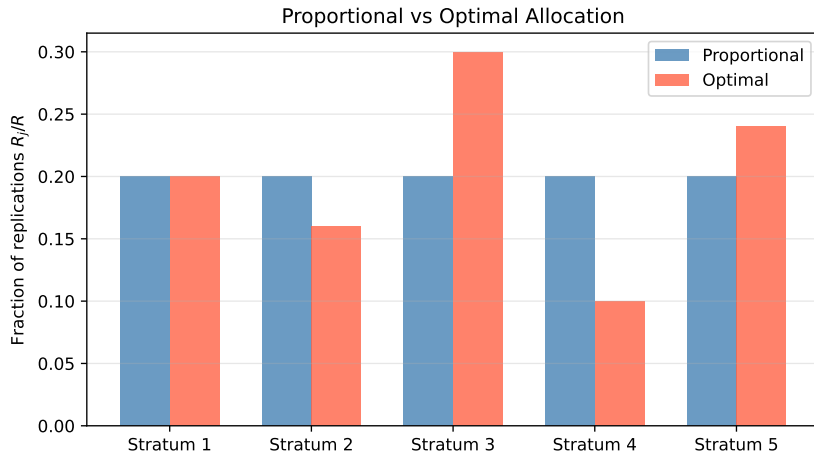
# Comparison: CMC vs stratified (proportional and optimal)
np.random.seed(42)
```

5.1.1 Stratified Sampling — Variance Comparison



This plot shows the variance (or standard error) of three estimators as a function of the number of strata m . Stratified sampling reduces variance significantly compared to CMC — the gap widens as m (number of strata) grows. Optimal allocation (assigning more samples to more variable strata) beats proportional allocation, which in turn beats CMC. All three converge to π as $R \rightarrow \infty$, but stratified estimators arrive at a given precision with

5.1.1 Optimal vs. Proportional Allocation



This bar chart shows the number of replications allocated to each of the $m = 10$ strata under optimal (dark bars) versus proportional (light bars) allocation. Under optimal allocation, the rightmost stratum receives far more replications because its within-stratum standard deviation σ_j is largest (the curve $4\sqrt{1-u^2}$ is steepest

1) Proportional allocation is not efficient, with $R_j/R = 1000$ for all strata. "14 / 109"

5.1.2 Antithetic Variates — Exploiting Negative Correlation I

The big idea. Instead of drawing R independent samples and averaging them, antithetic variates generate samples in *pairs* that are deliberately designed to be negatively correlated. When one sample overestimates I , its antithetic partner tends to underestimate I , so that their average cancels out the individual errors. This requires essentially no extra computation — you generate one pair instead of one independent sample.

5.1.2 Antithetic Variates — Exploiting Negative Correlation II

The Antithetic Estimator

Generate $R/2$ independent pairs $(Y_1^*, Y_1^{*'}), (Y_2^*, Y_2^{*'}), \dots, (Y_{R/2}^*, Y_{R/2}^{*'})$ such that each of Y_i^* and $Y_i^{*'}$ individually has the same distribution as Y , but within each pair the two are negatively correlated. Pairs are independent of one another.

The antithetic estimator is the average of all R values:

$$\hat{Y}_R^{\text{anti}} = \frac{1}{R/2} \sum_{i=1}^{R/2} \frac{Y_i^* + Y_i^{*'}}{2}.$$

Its variance compared to CMC is:

$$\text{Var}(\hat{Y}_R^{\text{anti}}) = \text{Var}(\hat{Y}_R^{\text{CMC}}) \cdot (1 + \text{Corr}(Y_1^*, Y_1^{*'})),$$

where Corr denotes the *correlation* (pronounced “core-el-AY-shun”), a number between -1 and $+1$ measuring the linear association between the two members of a pair. If this correlation is negative, the factor $(1 + \text{Corr})$ is less than 1, and variance decreases.

5.1.2 Antithetic Variates — Exploiting Negative Correlation III

How to Read the Variance Formula

The variance formula $\text{Var}(\hat{Y}_R^{\text{anti}}) = \text{Var}(\hat{Y}_R^{\text{CMC}}) \cdot (1 + \rho)$ (where ρ (rho) = $\text{Corr}(Y_i^*, Y_i^{*'})$) tells us directly how much we gain:

- If $\rho = -1$ (perfect negative correlation), the variance drops to zero — every pair averages exactly to I .
- If $\rho = 0$ (independent), we get the same variance as CMC — no gain, no loss.
- If $\rho > 0$ (positively correlated), variance actually *increases* compared to CMC — antithetics would be counterproductive.

The lesson: antithetics only help when the two members of a pair are negatively correlated. For monotone functions, this is always guaranteed.

5.1.2 Antithetic Variates — Exploiting Negative Correlation IV

Standard Construction: Complement in $[0, 1]$

If $Y = k(U_1, \dots, U_n)$ where $U_i \stackrel{\text{iid}}{\sim} \mathcal{U}[0, 1]$ (each U_i uniform on $[0, 1]$), define the antithetic pair via complementing each uniform:

$$Y^{\text{anti}} = k(1 - U_1, 1 - U_2, \dots, 1 - U_n).$$

Since U_i and $1 - U_i$ both have the $\mathcal{U}[0, 1]$ distribution, Y^{anti} has the same marginal distribution as Y . Moreover, if U_i is large, $1 - U_i$ is small, so the two outputs tend to pull in opposite directions — giving negative correlation.

5.1.2 Antithetic Variates — Exploiting Negative Correlation V

When Antithetics Are Guaranteed to Work (Theorem 5.1.13 & Corollary 5.1.14)

If $k(x_1, \dots, x_n)$ is *monotone* in each variable (meaning it is either non-decreasing or non-increasing in x_i when all other x_j are fixed), then:

$$\text{Cov}(k(U_1, \dots, U_n), k(1 - U_1, \dots, 1 - U_n)) \leq 0.$$

In words: for a monotone function, the complement construction always produces a non-positive covariance, which guarantees a variance reduction. The proof uses the inequality: if both k_1 and k_2 are monotone in the same direction, then $\mathbb{E}[k_1(X)k_2(X)] \geq \mathbb{E}[k_1(X)]\mathbb{E}[k_2(X)]$.

5.1.2 Antithetic Variates — Exploiting Negative Correlation VI

Worked Example — Estimating $\mathbb{E}[e^U]$ with Antithetics

Let $Y = e^U$ (e to the power U) where $U \sim \mathcal{U}[0, 1]$. The function $k(u) = e^u$ is strictly increasing in u , so Theorem 5.1.13 guarantees that the antithetic construction works.

The antithetic partner is $Y^{\text{anti}} = e^{1-U}$. The true value is $I = \mathbb{E}[e^U] = e - 1 \approx 1.71828$ (where $e \approx 2.71828$ is Euler's number).

CMC variance: $\text{Var}(e^U) = \mathbb{E}[e^{2U}] - (\mathbb{E}[e^U])^2 = \frac{e^2 - 1}{2} - (e - 1)^2 \approx 0.242$. (Here $\mathbb{E}[e^{2U}] = \int_0^1 e^{2u} du = (e^2 - 1)/2 \approx 3.195$.)

Antithetic average: $(e^U + e^{1-U})/2$. Its variance is:

$$\text{Var}\left(\frac{e^U + e^{1-U}}{2}\right) \approx 0.00474.$$

Variance reduction factor: $0.242/0.00474 \approx 51$. Using antithetic pairs gives a 51-fold reduction in variance.

Why so dramatic? When U is large (say $U = 0.9$), $e^{0.9} \approx 2.46$ overestimates I ; but $1 - U = 0.1$, so $e^{0.1} \approx 1.11$ underestimates I . The average $(2.46 + 1.11)/2 = 1.785$ is much closer to the truth 1.718.

5.1.2 Antithetic Variates — Python Code

```
import numpy as np

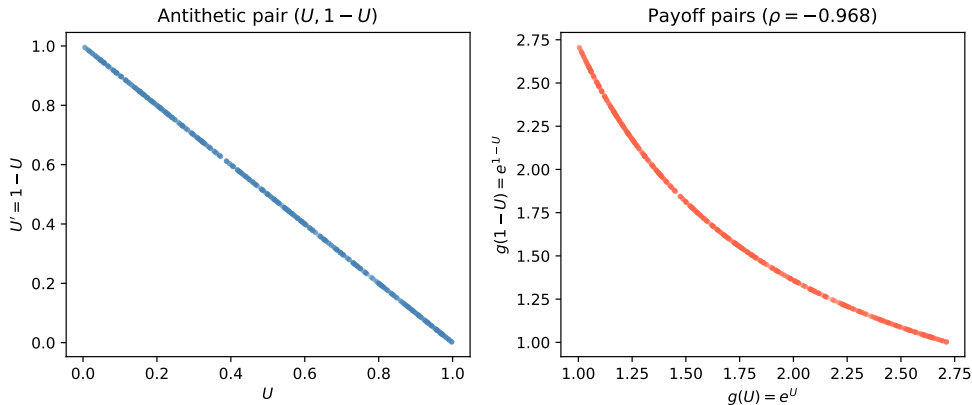
def antithetic_exp(R):
    """Estimate  $E[e^U]$ ,  $U \sim \text{Uniform}[0,1]$ , using antithetic variates."""
    U = np.random.uniform(0, 1, R // 2)
    Y1 = np.exp(U)           # original:  $e^U$ 
    Y2 = np.exp(1 - U)       # antithetic:  $e^{(1-U)}$ , the complement
    return np.mean((Y1 + Y2) / 2)

def antithetic_pi(R):
    """Estimate  $\pi$  using antithetic variates:  $Y = 4\sqrt{1-U^2}$ ."""
    U = np.random.uniform(0, 1, R // 2)
    Y1 = 4 * np.sqrt(1 - U**2)
    Y2 = 4 * np.sqrt(1 - (1 - U)**2)   # antithetic: use  $(1-U)$ 
    return np.mean((Y1 + Y2) / 2)

np.random.seed(42)
R = 100000
print(f"Antithetic  $E[e^U]$  = {antithetic_exp(R):.6f} (true: {np.e - 1:.6f})")
print(f"Antithetic  $\pi$  = {antithetic_pi(R):.6f} (true: {np.pi:.6f})")

# Variance comparison: run 2000 repetitions of size  $R=1000$ 
cmc_vars = np.var([np.mean(4 * np.sqrt(1 - np.random.uniform(0,1,1000)**2))
                    for _ in range(2000)])
anti_vars = np.var([antithetic_pi(1000) for _ in range(2000)])
print(f"CMC variance: {cmc_vars:.6f}")
print(f"Antithetic variance: {anti_vars:.6f}")
```

5.1.2 Antithetic Variates — Illustration



Left panel: The pair $(U, 1 - U)$ is perfectly negatively correlated — whenever U is large, $1 - U$ is small and vice versa.

Right panel: The corresponding payoffs $k(U)$ and $k(1 - U)$ also show strong negative correlation (for a monotone k), so their average $(k(U) + k(1 - U))/2$ has far lower variance than either one alone.

5.1.3 Common Random Numbers (CRN) — Comparing Two Systems I

The setting. Common random numbers (CRN) applies when the goal is not to estimate a single expectation $\mathbb{E}[Y]$, but rather the *difference* $\Delta = \mathbb{E}[Y_1] - \mathbb{E}[Y_2]$ between two system configurations. For example: “How much better is scheduling policy A than policy B?”

The naive approach draws independent samples from each system. CRN does something smarter: it uses the *same* random input stream to drive both simulations. If the same “random environment” (e.g., the same sequence of customer arrival times) is used for both systems, good runs tend to be good for both systems and bad runs bad for both — making Y_1 and Y_2 *positively* correlated. Since the variance of a difference $Y_1 - Y_2$ equals $\text{Var}(Y_1) + \text{Var}(Y_2) - 2\text{Cov}(Y_1, Y_2)$, positive covariance directly reduces the variance of our estimate of Δ .

5.1.3 Common Random Numbers (CRN) — Comparing Two Systems II

CRN Estimator

Use the same random stream U to simulate both systems in each replication. Generate R pairs $(Y_{1,j}, Y_{2,j})$, where both components use the same U_j :

$$\hat{\Delta}_R^{\text{CRN}} = \frac{1}{R} \sum_{j=1}^R (Y_{1,j} - Y_{2,j}).$$

The variance of this estimator is:

$$\text{Var}(\hat{\Delta}_R^{\text{CRN}}) = \frac{\text{Var}(Y_1) + \text{Var}(Y_2) - 2 \text{Cov}(Y_1, Y_2)}{R}.$$

CRN reduces variance whenever $\text{Cov}(Y_1, Y_2) > 0$ (positive covariance), which is the common case when both systems respond in the same direction to the shared random input.

5.1.3 Common Random Numbers (CRN) — Comparing Two Systems III

Why CRN induces positive covariance. If $Y_1 = k_1(U)$ and $Y_2 = k_2(U)$ are both non-decreasing (or both non-increasing) functions of the same input U , then by Theorem 5.1.13: $\text{Cov}(k_1(U), k_2(U)) \geq 0$. In plain language: when the shared random environment is favourable (a large U), both systems perform well. When it is unfavourable, both suffer. So Y_1 and Y_2 move together, and the difference $Y_1 - Y_2$ is much more stable than under independent replications.

Example 5.1.20 — Scheduling Policy Comparison

Consider two scheduling policies for a service queue: FCFS (First Come, First Served) versus LCFS (Last Come, First Served). We wish to estimate the difference in expected throughput Δ . Using CRN, the same sequence of service times and inter-arrival times is fed into both policies in every replication. When service times are long (a bad run for everyone), both FCFS and LCFS suffer similarly, so the difference $Y_1 - Y_2$ fluctuates little. Result reported in the book: for $R = 1,000$ replications, variance is reduced by a factor of approximately $22.5\times$ compared to using independent random streams for the two policies.

5.1.3 Common Random Numbers (CRN) — Comparing Two Systems IV

Synchronisation Warning

CRN requires careful “synchronisation” of the random number stream between the two systems. When the two systems have different structures (different numbers of events, different branching logic), a single shared stream may inadvertently map the same random number to logically different inputs in the two systems, destroying the positive correlation. Always design a clear mapping from the shared uniform stream to the same conceptual quantity in both systems.

5.1.4 Ratios of Expectations I

When the answer is itself a ratio. Sometimes the quantity of interest is $I = I^{(1)}/I^{(2)}$, a ratio of two unknown expectations $I^{(1)} = \mathbb{E}[Y^{(1)}]$ and $I^{(2)} = \mathbb{E}[Y^{(2)}]$. This arises naturally in several contexts: conditional expectations (where the numerator and denominator must both be estimated), and self-normalised importance sampling (Section 5.1.7), where the normalisation constant of a distribution is unknown.

5.1.4 Ratios of Expectations II

The Ratio Estimator

Generate R i.i.d. pairs $(Y_i^{(1)}, Y_i^{(2)})$ for $i = 1, \dots, R$ and define:

$$\hat{Y}_R^{\text{RoE}} = \frac{\sum_{i=1}^R Y_i^{(1)}}{\sum_{i=1}^R Y_i^{(2)}} = \frac{\hat{Y}_R^{(1)}}{\hat{Y}_R^{(2)}},$$

where $\hat{Y}_R^{(1)}$ ($\hat{Y}$ -hat superscript 1 sub R) is the sample mean of the numerator observations, and $\hat{Y}_R^{(2)}$ is the sample mean of the denominator observations.

Important: this estimator is *biased* — the ratio of sample means is not the same as the true ratio of population means. However, it is *consistent*: as $R \rightarrow \infty$, both sample means converge to their respective population means, so the ratio converges to $I^{(1)}/I^{(2)} = I$.

5.1.4 Ratios of Expectations III

Asymptotic Variance via the Delta Method

The *delta method* (also called the propagation of uncertainty formula) approximates the variance of a smooth function of sample means by linearising the function around its true value. For the ratio estimator, the asymptotic variance (the TAVC, times $1/R$) is:

$$(\varsigma^{\text{RoE}})^2 = \left(\frac{I^{(1)}}{I^{(2)}} \right)^2 \left[\frac{\Sigma_{\mathbf{Y}}(1, 1)}{(I^{(1)})^2} - \frac{2 \Sigma_{\mathbf{Y}}(1, 2)}{I^{(1)} I^{(2)}} + \frac{\Sigma_{\mathbf{Y}}(2, 2)}{(I^{(2)})^2} \right],$$

where $\Sigma_{\mathbf{Y}}(i, j) = \text{Cov}(Y^{(i)}, Y^{(j)})$ (the covariance matrix of the pair $(Y^{(1)}, Y^{(2)})$). In particular, $\Sigma_{\mathbf{Y}}(1, 1) = \text{Var}(Y^{(1)})$, $\Sigma_{\mathbf{Y}}(2, 2) = \text{Var}(Y^{(2)})$, and $\Sigma_{\mathbf{Y}}(1, 2) = \text{Cov}(Y^{(1)}, Y^{(2)})$.

The term $-2\Sigma_{\mathbf{Y}}(1, 2)$ appears in the variance formula with a minus sign: a positive covariance $\text{Cov}(Y^{(1)}, Y^{(2)}) > 0$ reduces the variance. This is the connection to CRN: by choosing the joint simulation of $(Y^{(1)}, Y^{(2)})$ to induce positive covariance, we can reduce the variance of the ratio estimator.

5.1.4 Ratios of Expectations IV

Application: Self-Normalised Importance Sampling

Self-normalised importance sampling (see Section 5.1.7) arises when the normalisation constant of the target distribution is unknown. The ratio estimator provides a consistent estimate, and the delta method supplies the confidence interval formula.

5.1.5 Control Variates — Using a Known Anchor I

The big idea. A *control variate* is an auxiliary random variable X that satisfies two conditions: (1) it is correlated with the output Y of interest, and (2) its population mean $\mu_X = \mathbb{E}[X]$ is *known exactly*. The trick is to correct the CMC estimate $\hat{Y}_R$ by subtracting the observed deviation of X from its known mean. If X and Y tend to move together, the correction will consistently remove the fluctuation — reducing variance.

5.1.5 Control Variates — Using a Known Anchor II

The Control Variate Estimator

Given primary output Y with unknown mean $I = \mathbb{E}[Y]$, control variate X with known mean $\mu_X = \mathbb{E}[X]$, and a scalar coefficient c (c , the control coefficient):

$$\hat{Y}_R^{\text{cv}} = \hat{Y}_R + c(\hat{X}_R - \mu_X),$$

where $\hat{Y}_R = \frac{1}{R} \sum_{i=1}^R Y_i$ and $\hat{X}_R = \frac{1}{R} \sum_{i=1}^R X_i$ are sample means from the same R replications (same runs produce both Y_i and X_i).

This estimator is *unbiased* because $\mathbb{E}[\hat{X}_R - \mu_X] = \mathbb{E}[\hat{X}_R] - \mu_X = 0$. Its variance is:

$$\text{Var}(\hat{Y}_R^{\text{cv}}) = \frac{1}{R} [\text{Var}(Y) + c^2 \text{Var}(X) + 2c \text{Cov}(X, Y)].$$

5.1.5 Control Variates — Using a Known Anchor III

Optimal Coefficient c^* and Resulting Variance

Minimising the variance over c (by differentiating with respect to c and setting the derivative to zero) gives:

$$c^* = -\frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

Here $\text{Cov}(X, Y)$ (covariance of X and Y , measuring how they move together) divided by $\text{Var}(X)$ is the ordinary regression slope of Y on X . The negative sign means that if $X > \mu_X$ (our run was lucky for X), we subtract a positive correction from $\hat{Y}_R$.

Substituting c^* back, the minimum variance is:

$$\text{Var}(\hat{Y}_R^{\text{cv},*}) = \frac{\text{Var}(Y)}{R} (1 - \rho^2),$$

where ρ (rho) = $\text{Corr}(X, Y)$ is the Pearson correlation coefficient between X and Y (a number between -1 and $+1$). The *variance reduction factor* is $1/(1 - \rho^2) \geq 1$.

5.1.5 Control Variates — Using a Known Anchor IV

How to Read the Variance Reduction Formula

The formula $(1 - \rho^2)$ tells the fraction of CMC variance that remains after using the control variate.

- If $|\rho| = 0.9$ (strong linear relationship), the factor is $1 - 0.81 = 0.19$, so the variance is reduced by about $5.3\times$ (equivalently, you need only $1/5.3 \approx 19\%$ as many simulations).
- If $|\rho| = 0$ (no linear relationship), the factor is 1 and there is no reduction — using an uncorrelated control variate wastes effort.
- If $|\rho| \rightarrow 1$ (perfect linear relationship), the variance approaches zero — the control variate “explains” all the variability in Y .

The control variate works by “explaining away” the variability in Y that is linearly predictable from X .

5.1.5 Control Variates — Using a Known Anchor V

Estimating c^* from a Pilot Simulation

In practice, $c^* = -\text{Cov}(X, Y)/\text{Var}(X)$ is unknown because the population covariance is not available. Run R' *pilot* replications and estimate:

$$\hat{c} = -\frac{\hat{S}_{XY}}{\hat{S}_X^2},$$

where $\hat{S}_{XY} = \frac{1}{R'-1} \sum_{i=1}^{R'} (X_i - \bar{X})(Y_i - \bar{Y})$ (the sample covariance, a weighted average of products of deviations from the respective sample means) and $\hat{S}_X^2 = \frac{1}{R'-1} \sum_i (X_i - \bar{X})^2$ (the sample variance of X). Using the estimated coefficient $\hat{c}$, the final estimator applied to R main replications is: $\hat{Y}_R^{\text{cv}} = \hat{Y}_R + \hat{c}(\hat{X}_R - \mu_X)$.

5.1.5 Control Variates — Using a Known Anchor VI

Example 5.1.24 — Estimating π with Control Variates

We estimate $I = \pi$ using $Y = 4\sqrt{1 - U^2}$, $U \sim \mathcal{U}[0, 1]$. Three control variates of increasing quality:

- **CV1:** $X = 1 - U$ with known mean $\mu_X = 1/2$. Correlation $\rho \approx 0.528$, giving variance reduction factor $\approx 1/(1 - 0.528^2) = 1/(1 - 0.279) \approx 1.4\times$.
- **CV2:** $X = \mathbf{1}(U_1 + U_2 > \sqrt{2})$, with μ_X computable. Higher correlation ρ , variance reduction $\approx 4\times$.
- **CV3:** A degree-4 polynomial in U , calibrated to approximate Y . Very high correlation: $1 - \rho^2 = 0.0000860$, giving an astounding variance reduction of $\approx 11,630\times$.

The lesson: a cleverly chosen control variate can achieve enormous reductions. The more tightly X and Y are correlated (higher $|\rho|$), the better.

5.1.5 Control Variates — Python Code

```
import numpy as np

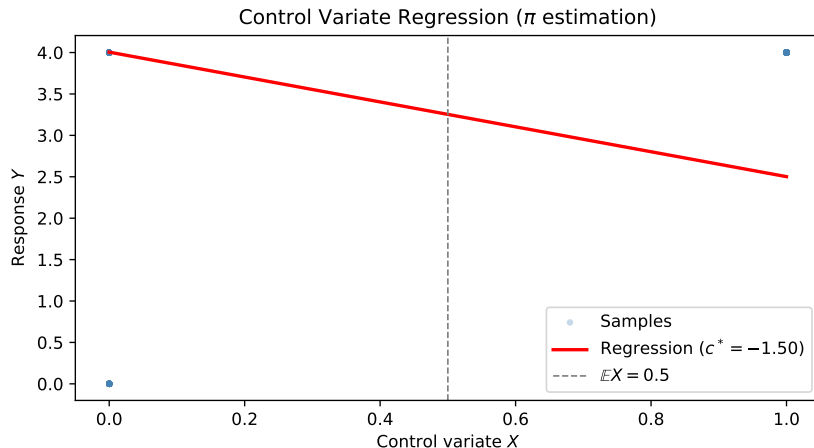
def control_variate_pi(R, R_pilot=1000):
    """Estimate pi using control variate  $X = 1-U$ , with  $E[X] = 0.5$ ."""
    # Pilot phase: estimate  $c^* = -\text{Cov}(X, Y) / \text{Var}(X)$ 
    U_p = np.random.uniform(0, 1, R_pilot)
    Y_p = 4 * np.sqrt(1 - U_p**2)    # primary output  $Y = 4\sqrt{1-U^2}$ 
    X_p = 1 - U_p                    # control variate  $X = 1-U$ 

    cov_mat = np.cov(X_p, Y_p, ddof=1)    # 2x2 covariance matrix
    c_hat = -cov_mat[0, 1] / cov_mat[0, 0] # estimated  $c^*$ 
    rho = np.corrcoef(X_p, Y_p)[0, 1]     # estimated correlation
    print(f"  Pilot estimate:  $c^* = \{c\_hat:.4f\}$ ,  $\rho = \{rho:.4f\}$ ")
    print(f"  Theoretical var. reduction =  $\{1/(1 - rho**2):.2f\}x$ ")

    # Main simulation phase: apply control variate correction
    U = np.random.uniform(0, 1, R)
    Y = 4 * np.sqrt(1 - U**2)
    X = 1 - U
    Y_cv = Y + c_hat * (X - 0.5)    # subtract  $c\_hat * (X\_bar - \mu_X)$ 
    est = np.mean(Y_cv)
    se = np.std(Y_cv, ddof=1) / np.sqrt(R)
    return est, se

np.random.seed(42)
est, se = control_variate_pi(R=1000000)
print(f"CV estimate:  $\{est:.6f\} \pm \{1.96 * se:.6f\}$  (95% CI)")
```

5.1.5 Control Variates — The Regression Interpretation



The control variate estimator can be understood geometrically as a regression adjustment. In this scatter plot, each point is one simulation run plotted as (X_i, Y_i) . The regression line through these points has slope $-c^* = \text{Cov}(X, Y)/\text{Var}(X)$. The control variate correction $c^*(\hat{X}_R - \mu_X)$ moves the estimate along this regression line to the point where $Y = \mu_Y$. The residuals from this regression (i.e., how far each Y_i lies above or

5.1.6 Conditional Monte Carlo — The Rao-Blackwell Idea I

The idea. Sometimes the output Y can be decomposed into two parts: an observable quantity X and remaining randomness that is conditionally independent of X . If we can compute the conditional expectation $c(x) = \mathbb{E}[Y \mid X = x]$ analytically (or at negligible cost), then we can replace the noisy individual sample Y with the smooth function $c(X)$. By the law of total expectation, $\mathbb{E}[c(X)] = \mathbb{E}[Y] = I$, so $c(X)$ is an unbiased estimator of I — but with lower variance.

The Conditional Monte Carlo Estimator

Let $c(x) = \mathbb{E}[Y \mid X = x]$ (c of x , the conditional mean of Y given that the auxiliary variable X equals x). Generate $X_1, X_2, \dots, X_R$ as i.i.d. copies of X and compute:

$$\hat{Y}_R^{\text{cond}} = \frac{1}{R} \sum_{j=1}^R c(X_j).$$

Each term $c(X_j)$ is the analytically-computed conditional mean, not a single noisy sample — so the estimator is smoother.

5.1.6 Conditional Monte Carlo — The Rao-Blackwell Idea II

The Rao-Blackwell Theorem: Conditioning Always Reduces Variance

By the law of total variance:

$$\text{Var}(Y) = \underbrace{\mathbb{E}[\text{Var}(Y \mid X)]}_{\geq 0} + \underbrace{\text{Var}(\mathbb{E}[Y \mid X])}_{=\text{Var}(c(X))}.$$

Rearranging: $\text{Var}(c(X)) = \text{Var}(Y) - \mathbb{E}[\text{Var}(Y \mid X)]$. Since $\mathbb{E}[\text{Var}(Y \mid X)] \geq 0$ (it is an average of non-negative quantities), we conclude:

$$\text{Var}(c(X)) \leq \text{Var}(Y),$$

with strict inequality unless Y is already a deterministic function of X (i.e., no residual randomness after observing X).

In words: replacing the noisy Y with the smooth conditional expectation $c(X) = \mathbb{E}[Y \mid X]$ always reduces (or maintains) variance. The more of Y 's variability that is explained by X , the greater the reduction.

5.1.6 Conditional Monte Carlo — The Rao-Blackwell Idea III

Example 5.1.29 — Estimating π via Conditional MC

Consider estimating π via the quarter-circle indicator: $Y = 4 \cdot \mathbf{1}(U_1^2 + U_2^2 \leq 1)$, where $U_1, U_2 \stackrel{\text{iid}}{\sim} \mathcal{U}[0, 1]$. Here Y is a Bernoulli-type indicator (it equals 4 if the point (U_1, U_2) falls inside the quarter-circle, and 0 otherwise), which is intrinsically noisy.

Condition on the first coordinate $X = U_1$:

$$c(x) = \mathbb{E}[Y \mid U_1 = x] = 4 \cdot \mathbb{P}(U_2 \leq \sqrt{1 - x^2}) = 4\sqrt{1 - x^2}.$$

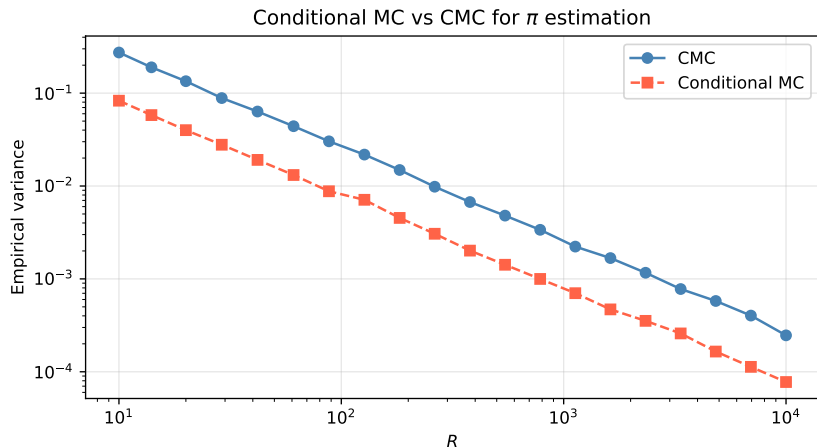
In words: given that $U_1 = x$, the probability that the second coordinate U_2 falls below $\sqrt{1 - x^2}$ (the height of the unit circle at x) is exactly $\sqrt{1 - x^2}$, since $U_2 \sim \mathcal{U}[0, 1]$.

The conditional MC estimator is: $\hat{Y}_R^{\text{cond}} = \frac{4}{R} \sum_{i=1}^R \sqrt{1 - U_i^2}$.

Variance comparison:

- $\text{Var}(\hat{Y}_R^{\text{CMC}}) = 2.697/R$ (indicator $Y \in \{0, 4\}$ is very noisy).
- $\text{Var}(\hat{Y}_R^{\text{cond}}) = 0.797/R$ (smooth function $4\sqrt{1 - U^2}$ is much less variable).
- Variance reduction factor: $2.697/0.797 \approx \mathbf{3.38\times}$.

5.1.6 Conditional MC — Variance Comparison



The conditional MC estimator replaces the binary indicator $4 \cdot \mathbf{1}(U_1^2 + U_2^2 \leq 1)$ (which is extremely noisy: it is either 0 or 4) with the smooth, deterministic function $c(U_1) = 4\sqrt{1 - U_1^2}$ (which is always between 0 and 4 and varies continuously). Both estimators converge at rate $1/\sqrt{R}$, but the conditional estimator does so with a much smaller constant in front.

5.1.7 Importance Sampling — The Change-of-Measure Idea I

Motivation. Crude Monte Carlo estimating $I = \mathbb{E}_f[k(X)] = \int k(x)f(x) dx$ by sampling from the density f (f, the original “target” density) works well when $k(x)$ is non-negligible over most of the support of f . But if $k(x)$ is large only in a rare region where $f(x)$ is tiny — for example, when estimating the probability of a rare event — most samples from f give $k(X) \approx 0$ and contribute nothing useful.

Importance sampling (IS) solves this by sampling from a different density $\tilde{f}$ (f-tilde, the *proposal* or *importance* density) that concentrates samples where $k(x)$ is large. To correct for the change in distribution, each sample is reweighted by the ratio of the old density to the new one.

5.1.7 Importance Sampling — The Change-of-Measure Idea II

The Importance Sampling Identity

Let $\tilde{f}$ (f-tilde) satisfy $\tilde{f}(x) > 0$ wherever $k(x)f(x) \neq 0$. Define the *likelihood ratio* $L(x)$ (also called the importance weight):

$$L(x) = \frac{f(x)}{\tilde{f}(x)}.$$

Then the integral I can be rewritten as an expectation under $\tilde{f}$:

$$I = \int_E k(x)f(x) dx = \int_E k(x) \frac{f(x)}{\tilde{f}(x)} \tilde{f}(x) dx = \mathbb{E}_{\tilde{f}}[k(\tilde{X}) \cdot L(\tilde{X})],$$

where $\tilde{X} \sim \tilde{f}$ (tilde-X is a random variable drawn from the proposal density $\tilde{f}$). We sample $\tilde{X}$ from the new distribution and multiply by the weight $L(\tilde{X})$ to compensate.

5.1.7 Importance Sampling — The Change-of-Measure Idea III

The IS Estimator

Generate $\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_R \stackrel{\text{iid}}{\sim} \tilde{f}$ and compute:

$$\hat{Y}_R^{\text{IS}} = \frac{1}{R} \sum_{j=1}^R k(\tilde{X}_j) \frac{f(\tilde{X}_j)}{\tilde{f}(\tilde{X}_j)}.$$

Each sample $\tilde{X}_j$ (tilde-X sub j) is drawn from the proposal, and $L(\tilde{X}_j) = f(\tilde{X}_j)/\tilde{f}(\tilde{X}_j)$ (L of tilde-X sub j) is the likelihood ratio weight that corrects for having sampled from $\tilde{f}$ rather than f . By the identity above, this estimator is unbiased: $\mathbb{E}_{\tilde{f}}[\hat{Y}_R^{\text{IS}}] = I$.

5.1.7 Importance Sampling — The Change-of-Measure Idea IV

Variance of the IS Estimator (Proposition 5.1.33)

By the central limit theorem, the IS estimator satisfies:

$$\sqrt{R}(\hat{Y}_R^{\text{IS}} - I) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \varsigma^2),$$

where the asymptotic variance ς^2 (varsigma squared) is:

$$\varsigma^2 = \int_E \left(k(x) \frac{f(x)}{\tilde{f}(x)} \right)^2 \tilde{f}(x) dx - I^2 = \text{Var}_{\tilde{f}} \left[k(\tilde{X}) L(\tilde{X}) \right].$$

To reduce variance, we want the weighted integrand $k(x)f(x)/\tilde{f}(x)$ to be as *flat* (constant) as possible across the support of $\tilde{f}$. If it is perfectly constant, the variance is zero.

5.1.7 Importance Sampling — The Change-of-Measure Idea V

The Optimal Proposal Distribution (Zero-Variance IS)

The variance ς^2 is minimised — in fact, equals zero — when the proposal is chosen proportional to the absolute integrand:

$$\tilde{f}^*(x) = \frac{|k(x)| f(x)}{\int_E |k(x)| f(x) dx} = \frac{|k(x)| f(x)}{I'},$$

where $I' = \int_E |k(x)| f(x) dx$ is the normalisation constant. Under this proposal, $k(x)f(x)/\tilde{f}^*(x) = I'$ is exactly constant, so the IS estimator has zero variance — every sample gives exactly I' .

Caveat: Computing I' requires knowing the integral, which is essentially the same as knowing I itself. So this zero-variance proposal cannot be used directly. It serves instead as a *target*: the closer $\tilde{f}$ is to $\tilde{f}^*$, the lower the variance. The practical guideline is to choose $\tilde{f}$ with a shape similar to $|k(x)|f(x)$ — concentrating samples where the integrand is large.

5.1.7 Importance Sampling — The Change-of-Measure Idea VI

Example 5.1.34 — IS for a Cauchy Tail Probability

Estimate $I = \mathbb{P}(X > 2)$ where X has a Cauchy distribution with density $f(x) = 1/(\pi(1 + x^2))$ (pi in the denominator is the normalisation constant for the Cauchy density, not the number we are estimating). The true value is $I = 1/2 - \arctan(2)/\pi \approx 0.14758$.

CMC: Draw $X_i \sim \text{Cauchy}$ and compute $Y_i = \mathbf{1}(X_i > 2)$. Variance:
 $\text{Var}(\hat{Y}_R^{\text{CMC}}) = I(1 - I)/R = 0.1258/R$.

IS with a Pareto-type proposal concentrated on $(2, \infty)$: $\tilde{f}(x) = (2/x^2)\mathbf{1}(x > 2)$. To simulate $\tilde{X} \sim \tilde{f}$, use the inverse CDF: $\tilde{X} = 2/U$ where $U \sim \mathcal{U}[0, 1]$. The likelihood ratio is $L(\tilde{X}) = f(\tilde{X})/\tilde{f}(\tilde{X})$.

IS variance: $\text{Var}(\hat{Y}_R^{\text{IS}}) = 9.55 \times 10^{-5}/R$.

Reduction factor: $0.1258/(9.55 \times 10^{-5}) \approx \mathbf{1317\times}$.

IS works so dramatically here because CMC wastes most samples on $X < 2$ where $k(X) = \mathbf{1}(X > 2) = 0$. The IS proposal puts every sample above 2, exactly where k is non-zero.

5.1.7 IS for Rare Events — Shifting the Normal Distribution I

5.1.7 IS for Rare Events — Shifting the Normal Distribution II

Example 5.1.36 — IS for $\mathbb{P}(Z > 4)$, $Z \sim \mathcal{N}(0, 1)$

The standard normal tail probability $I = \mathbb{P}(Z > 4)$ is very small: $I \approx 3.167 \times 10^{-5}$ (roughly 3 in 100,000). This is a classic *rare event*: CMC draws $Z \sim \mathcal{N}(0, 1)$ and checks whether $Z > 4$, but with probability $\approx 99.997\%$ the answer is “no” — almost every sample contributes nothing. With $R = 10^5$ runs, CMC typically sees 0, 1, 2 or 3 successes, making accurate estimation essentially impossible.

IS proposal: Use a shifted normal $\tilde{f} = \mathcal{N}(\theta, 1)$ (a normal distribution with mean θ (theta) shifted toward the rare-event region $Z > 4$). The likelihood ratio (the ratio of the standard normal density to the shifted normal density) is:

$$L(z) = \frac{\phi_{0,1}(z)}{\phi_{\theta,1}(z)} = \exp\left(-\theta z + \frac{\theta^2}{2}\right),$$

where $\phi_{0,1}(z)$ (phi sub 0-1 of z) is the standard normal density $\phi_{0,1}(z) = (2\pi)^{-1/2} \exp(-z^2/2)$, and $\phi_{\theta,1}(z)$ (phi sub theta-1) is the density of $\mathcal{N}(\theta, 1)$.

The IS estimator becomes:

$$\hat{Y}_R^{\text{IS}, \theta} = \frac{1}{R} \sum_{i=1}^R \mathbf{1}(\tilde{Z}_i > 4) \exp\left(-\theta \tilde{Z}_i + \frac{\theta^2}{2}\right), \quad \tilde{Z}_i \sim \mathcal{N}(\theta, 1).$$

5.1.7 Importance Sampling — Python Code

```
import numpy as np
from scipy.stats import norm

def IS_normal_tail(theta, R=100000):
    """Estimate  $P(Z > 4)$ ,  $Z \sim N(0,1)$ , using IS with  $N(\theta, 1)$  proposal."""
    Z = np.random.normal(theta, 1, R)
    # Log likelihood ratio:  $\log[\phi(z; 0,1) / \phi(z; \theta, 1)]$ 
    # =  $-\theta z + \theta^2/2$ 
    log_LR = -theta * Z + theta**2 / 2
    Y_IS = (Z > 4).astype(float) * np.exp(log_LR)
    est = np.mean(Y_IS)
    var = np.var(Y_IS, ddof=1) / R
    return est, var

np.random.seed(42)
R = 100000
true_I = 1 - norm.cdf(4) # approximately 3.167e-5
print(f"True  $P(Z > 4) = \{true\_I:.4e\}")$ 
print()

for theta in [0, 2, 4, 4.2256]:
    est, var = IS_normal_tail(theta=theta, R=R)
    cmc_var = true_I * (1 - true_I) / R # CMC variance for reference
    reduction = (true_I * (1 - true_I)) / (var * R) if var > 0 else float('inf')
    label = "(CMC)" if theta == 0 else ""
    print(f"theta = {theta:.4f} {label:6s}: "
          f"est = {est:.3e}, var/R = {var:.3e}, "
```

5.1.7 Self-Normalised Importance Sampling I

When normalisation constants are unknown. In Bayesian statistics and many machine learning applications, the target density f is only known up to a multiplicative constant — for example, $f(x) \propto f_0(x)$ where f_0 is computable but the normalisation $\int f_0(x) dx$ is not. Similarly, the proposal $\tilde{f}$ may only be known up to a constant. In this case, the standard IS estimator cannot be computed because $L(x) = f(x)/\tilde{f}(x)$ requires the exact densities. Self-normalised IS (SNIS) handles this by forming a ratio that cancels the unknown constants.

5.1.7 Self-Normalised Importance Sampling II

Self-Normalised IS Estimator

Write $f(x) = c \cdot f_0(x)$ and $\tilde{f}(x) = \tilde{c} \cdot \tilde{f}_0(x)$ where the normalisation constants c and $\tilde{c}$ are unknown. Define *unnormalised weights* $w(x) = f_0(x)/\tilde{f}_0(x)$ (the ratio of the known parts of the densities). Then it can be shown that:

$$I = \mathbb{E}_f[k(X)] = \frac{\mathbb{E}_{\tilde{f}}[k(\tilde{X}) w(\tilde{X})]}{\mathbb{E}_{\tilde{f}}[w(\tilde{X})]}.$$

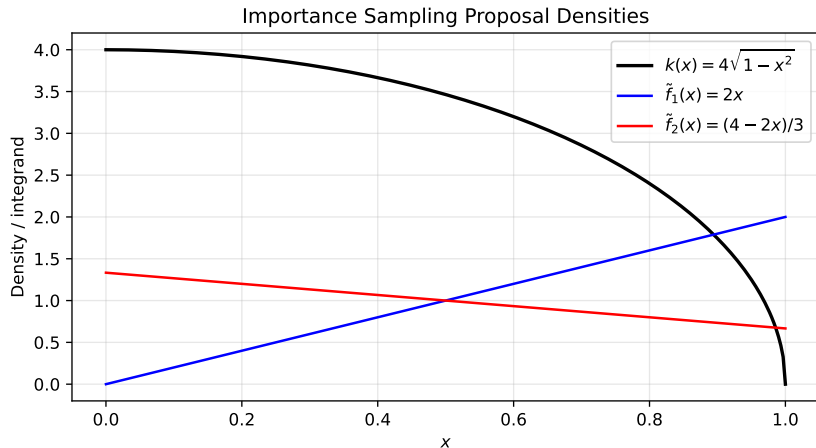
The SNIS estimator replaces numerator and denominator by their respective sample means:

$$\hat{Y}_R^{\text{SN}} = \frac{\sum_{i=1}^R k(\tilde{X}_i) w(\tilde{X}_i)}{\sum_{i=1}^R w(\tilde{X}_i)} = \sum_{i=1}^R k(\tilde{X}_i) w_i^{\text{norm}},$$

where $w_i^{\text{norm}} = w(\tilde{X}_i) / \sum_{j=1}^R w(\tilde{X}_j)$ (w sub i, normalised) are the *normalised weights* — each lies between 0 and 1 and all sum to 1. The unknown constants c and $\tilde{c}$ cancel.

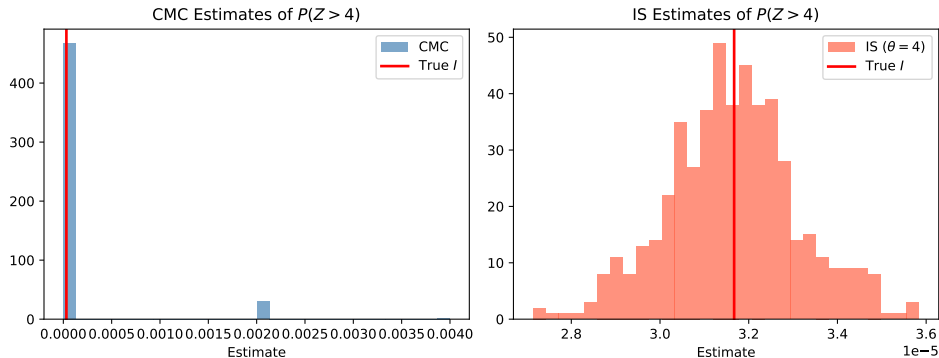
This estimator is *biased* (the ratio of sample means is not the same as the ratio of population means),

5.1.7 IS Proposal Distributions — Choosing Wisely



This figure compares the integrand $k(x)f(x)$ (the “target shape”) with several candidate proposals $\tilde{f}$. The ideal proposal $\tilde{f}^*(x)$ is proportional to $|k(x)|f(x)$, making the IS weight $L(x) = f(x)/\tilde{f}(x)$ constant. Among practical proposals, the one whose shape best matches $k(x)f(x)$ achieves the lowest variance. A proposal with too light tails (missing some region where the integrand is large) inflates variance catastrophically; too heavy

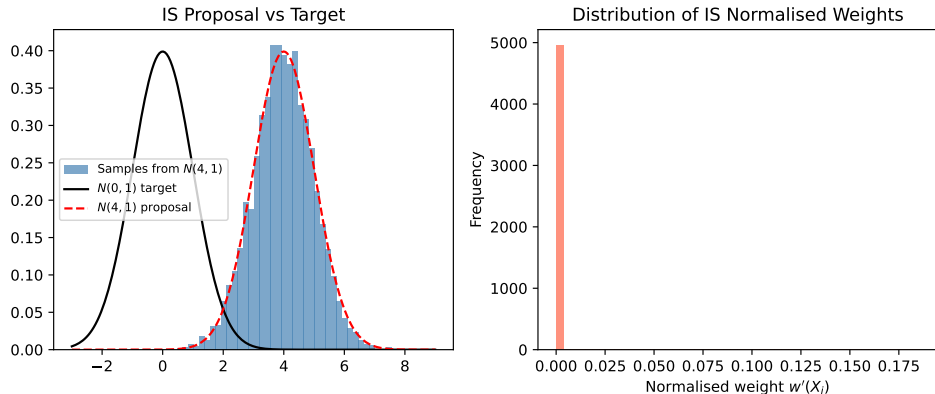
5.1.7 IS for Rare Events: Estimating $\mathbb{P}(Z > 4)$



Top panel (CMC): Most estimates are exactly 0 because the event $Z > 4$ almost never occurs in standard normal draws (probability $\approx 0.003\%$). Estimates are highly unstable and many runs of $R = 10^4$ produce zero hits.

Bottom panel (IS with $\tilde{f} = \mathcal{N}(4, 1)$): Every sample is drawn near the rare-event region. About half fall above 4 and contribute non-zero, well-calibrated weights. Estimates are stable and tightly clustered around the true value.

5.1.7 Self-Normalised IS — Weight Distribution



This histogram shows the distribution of normalised IS weights $w_i^{\text{norm}} = w(\tilde{X}_i) / \sum_j w(\tilde{X}_j)$. For a good proposal, the weights are roughly equal (each sample contributes about $1/R$ to the estimate) — no single sample dominates. For a poor proposal, one or two samples receive almost all the weight, making the estimate effectively based on a single observation (very unstable). A well-spread weight distribution is a diagnostic sign that the IS proposal is doing its job.

5.1.8 The Cross-Entropy Method — Finding the Best Proposal I

The problem IS leaves unsolved. Importance sampling is only as good as the proposal distribution $\tilde{f}$. For simple problems, the ideal proposal shape can be guessed (e.g., shift a normal distribution to the rare-event boundary). But for complex problems, the optimal proposal is unknown — and a poor choice can actually increase variance compared to CMC.

The *cross-entropy (CE) method* provides a principled, iterative procedure to find the best proposal from a parametric family $\{\tilde{f}_\theta : \theta \in \Theta\}$ (a family of candidate proposals indexed by a parameter θ , theta). Instead of selecting θ by hand, CE adapts it automatically.

5.1.8 The Cross-Entropy Method — Finding the Best Proposal II

KL Divergence and the CE Objective

The *Kullback-Leibler (KL) divergence* (also called cross-entropy loss or relative entropy) from the zero-variance proposal $g^*(x) = k(x)f(x)/I$ to the parametric proposal $\tilde{f}_\theta$ is:

$$D_{\text{KL}}(g^* \parallel \tilde{f}_\theta) = \int g^*(x) \log \frac{g^*(x)}{\tilde{f}_\theta(x)} dx.$$

Here D_{KL} (D sub KL) measures how different $\tilde{f}_\theta$ is from the ideal proposal g^* : larger values mean more discrepancy. Minimising D_{KL} over θ is equivalent to maximising:

$$\theta^* = \arg \max_{\theta \in \Theta} \mathbb{E}_f \left[k(X) \log \tilde{f}_\theta(X) \right].$$

This can be estimated from M samples $X_1, \dots, X_M \sim f$ as:

$$\hat{\theta}^* = \arg \max_{\theta \in \Theta} \frac{1}{M} \sum_{i=1}^M k(X_i) \log \tilde{f}_\theta(X_i).$$

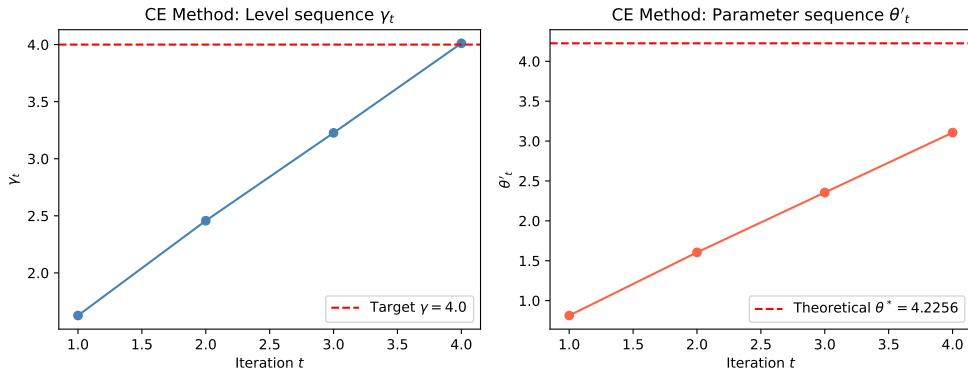
For many standard families (normal, exponential, Bernoulli), this maximisation has a closed-form

5.1.8 Cross-Entropy Method — Python Code

```
import numpy as np
from scipy.stats import norm

def CE_normal_1D(gamma=4.0, M=5000, rho=0.1, alpha=0.7, max_iter=100):
    """
    CE method to find optimal IS proposal mean for  $P(Z > \text{gamma})$ ,
     $Z \sim N(0, 1)$ , within the parametric family  $\{N(\mu, 1)\}$ .
    """
    mu = 0.0 # start from the original distribution  $N(0, 1)$ 
    history = []
    for t in range(max_iter):
        X = np.random.normal(mu, 1, M)
        # Likelihood ratio weights:  $\phi(x; 0, 1) / \phi(x; \mu, 1)$ 
        # Using log form for numerical stability:  $\log LR = -\mu * x + \mu^2 / 2$ 
        w = np.exp(-mu * X + mu**2 / 2)
        # Current level:  $(1 - \rho)$ -quantile of performance values  $h(X) = X$ 
        gamma_t = np.quantile(X, 1 - rho)
        # Update mu: weighted mean of elite samples (those  $\geq \text{gamma}_t$ )
        mask = X >= gamma_t
        if mask.sum() > 0:
            new_mu = np.sum(w[mask] * X[mask]) / np.sum(w[mask])
            mu = alpha * new_mu + (1 - alpha) * mu # smoothed update
        history.append({'iter': t+1, 'gamma_t': gamma_t, 'mu': mu})
        if gamma_t >= gamma:
            break
    return mu, history
```

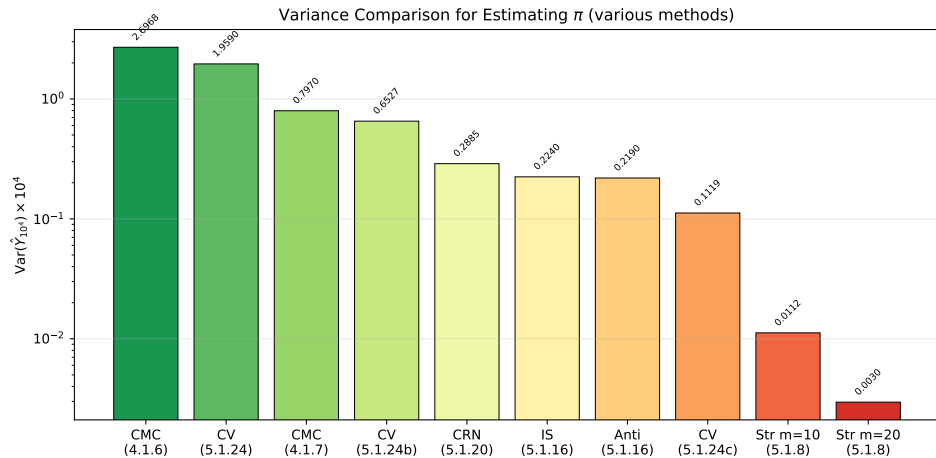
5.1.8 Cross-Entropy Method — Convergence



Left panel: The intermediate level γ_t (gamma sub t , the current performance threshold) climbs from near 0 toward the target $\gamma = 4$ over approximately 46 iterations. At each step, about 10% of the current samples exceed γ_t (the “elite” samples), pulling the next level up.

Right panel: The proposal mean μ'_t (mu sub t prime) converges to the CE-optimal value $\mu^* = 4.2256$. The smoothing factor $\alpha = 0.7$ gives a smooth, monotone convergence without oscillation.

5.1.9 Summary — Variance Comparison Across All Methods



This figure plots the variance of each estimator on a logarithmic (log) scale as a function of replications R , all targeting $I = \pi$. All lines are straight (because variance $\propto 1/R$ on a log-log plot), but they differ in vertical position — each method's line lies below the CMC baseline by its variance reduction factor.

5.1.9 Summary Table — All Variance Reduction Methods for π I

The following table summarises every variance reduction technique covered in Section 5.1, all applied to estimating $I = \pi$ using $R = 10^4$ replications. The variances are scaled by 10^4 for readability; the “Reduction” column gives the ratio of each method’s variance to CMC.

Method	Estimator / description	Var $\times 10^4$	Reduction
CMC	$Y = 4\mathbf{1}(U_1^2 + U_2^2 \leq 1)$ (indicator)	2.697	1 $\times$
Cond	$Y = 4\sqrt{1 - U^2}$ (conditional on U_1)	0.797	3.4 $\times$
CV1	$X = \mathbf{1}(U_1 + U_2 \leq 1)$	1.959	1.4 $\times$
CV2	$X = \mathbf{1}(U_1 + U_2 \geq \sqrt{2})$	0.653	4.1 $\times$
Anti	$Y_1 = 4\sqrt{1 - U^2}$, $Y_2 = 4\sqrt{1 - (1 - U)^2}$	0.219	12 $\times$
CRN	$Y_1 = 8/(1 + U^2) - 4\sqrt{1 - U^2}$	0.289	9.3 $\times$
IS	proposal $\tilde{f}(x) = (4 - 2x)/3$	0.224	12 $\times$
CV3	degree-4 polynomial control variate	0.112	24 $\times$
Str $m = 10$	10 strata, optimal allocation	0.0112	240 $\times$
Str $m = 20$	20 strata, optimal allocation	0.00296	911 $\times$

5.1.9 Summary Table — All Variance Reduction Methods for π II

Reading the table. The variance column shows how much the estimator's output fluctuates around the true value π . A smaller variance means that with the same $R = 10^4$ runs, our estimate is more reliable. Several lessons stand out:

- ① The *conditional MC* estimator (replacing the 0/4 indicator with the smooth function $4\sqrt{1 - U^2}$) gives a free $3.4\times$ reduction simply by computing a conditional expectation analytically.
- ② *Antithetic variates* and basic *importance sampling* achieve a moderate $12\times$ reduction with very simple implementations.
- ③ A *polynomial control variate* (CV3) achieves $24\times$ by fitting a high-degree polynomial to the integrand shape.
- ④ *Stratified sampling* with many strata dominates all other methods: $m = 20$ strata with optimal allocation gives $911\times$ improvement, meaning we need only about 11,000 stratified samples to match the precision of 10^7 CMC samples.

5.1.9 Summary Table — All Variance Reduction Methods for π III

Key Takeaway — Choosing the Right Method

Methods that exploit the *shape* of the integrand (polynomial control variates, stratification with many strata) achieve the greatest reductions. Antithetic variates, CRN, and basic IS give moderate but practical gains with simple implementations. In real applications, the best strategy often combines several methods — for example, stratified sampling with antithetic variates within each stratum.

5.2 Finance and Actuarial Science – Overview I

The methods introduced in Section 5.1 find their most dramatic applications in quantitative finance. Classical financial mathematics models the evolution of a stock price as a *geometric Brownian motion* (GBM), and the price of a financial option is then the discounted expected payoff under this model. For simple *European* options a closed-form answer exists (the Black-Scholes formula), but most real-world derivatives — Asian options, barrier options, basket options on many underlying assets — require numerical methods. Monte Carlo simulation is the standard tool, and variance reduction can cut computational cost by factors ranging from ten to tens of thousands.

Topics Covered in Section 5.2

- 1 Brownian motion: definition, key properties, and two simulation algorithms.
- 2 Financial contracts: European and Asian options, the Black-Scholes formula.
- 3 Stratified sampling applied to option pricing.
- 4 Control variates for the Asian option (geometric mean as control).
- 5 Antithetic variates for the Asian option.
- 6 Importance sampling for the European call option.

5.2 Finance and Actuarial Science – Overview II

Running Parameters (used in all examples unless stated otherwise)

Throughout Section 5.2 we fix: $r = 0.05$ (the annual risk-free interest rate, meaning \$1 invested risk-free today grows to $e^{0.05} \approx \$1.051$ after one year), σ (sigma) = 0.25 (volatility, measuring how wildly the stock price fluctuates per year), $S(0) = 100$ (the initial stock price in dollars), $K = 100$ (the strike price, i.e., the agreed purchase price in the option contract).

5.2.1 Brownian Motion – Definition and Properties I

Brownian motion is the fundamental continuous-time random process from which all GBM models are built. Think of it as the mathematical model for the path traced by a dust particle buffeted by random molecular collisions in a fluid: the path is continuous (no sudden jumps), but it is so erratic that it is not differentiable anywhere.

5.2.1 Brownian Motion – Definition and Properties II

Definition 5.2.1 – Standard Brownian Motion

A stochastic process $(B(t) : t \in [0, T])$ is a *standard Brownian motion* (abbreviated BM) if all four conditions hold:

- (i) $B(0) = 0$ (the process starts at zero at time $t = 0$).
- (ii) **Independent increments:** For any sequence of times $0 \leq t_1 < t_2 < \dots < t_n \leq T$, the differences $B(t_2) - B(t_1)$, $B(t_3) - B(t_2)$, $\dots$ are mutually independent random variables. Knowing what happened in one time-interval gives no information about what happens in any non-overlapping interval.
- (iii) **Normally distributed increments:** For any $0 \leq s \leq t$, the increment $B(t) - B(s) \sim \mathcal{N}(0, t - s)$. In words, the change in B over a time interval of length $\Delta t = t - s$ is normally distributed with mean 0 and variance Δt (so standard deviation $\sqrt{\Delta t}$).
- (iv) $t \mapsto B(t)$ is a continuous function of time (no jumps).

5.2.1 Brownian Motion – Definition and Properties III

Key Consequence: Covariance Formula (Proposition 5.2.2)

For any $0 \leq s \leq t$, the covariance (a measure of how much $B(s)$ and $B(t)$ move together) is:

$$\text{Cov}(B(s), B(t)) = \min(s, t) = s.$$

Because $B(t) = B(s) + [B(t) - B(s)]$ and the increment $B(t) - B(s)$ is independent of $B(s)$, we get $\text{Cov}(B(s), B(t)) = \text{Var}(B(s)) = s$.

The correlation (the covariance rescaled to lie in $[-1, 1]$) is:

$$\text{Corr}(B(s), B(t)) = \sqrt{\frac{s}{t}}.$$

This makes intuitive sense: observations close in time (large s/t) are highly correlated; observations far apart in time (small s/t) are nearly uncorrelated.

5.2.1 Brownian Motion – Definition and Properties IV

How to Read This

The variance of $B(t)$ equals t itself. This means the uncertainty about the BM path *grows linearly with time*: predicting a stock price further into the future is proportionally harder. The standard deviation is $\sqrt{t}$, which grows more slowly than t — a reflection of the famous “square root of time” rule used in finance.

5.2.1 Brownian Motion – Definition and Properties V

Simulation on a Uniform Grid (Algorithm 36)

To simulate BM on the uniform grid $t_i = i \Delta$, where $\Delta = T/N$ and $i = 0, 1, \dots, N$ (so there are $N + 1$ time points in $[0, T]$):

- 1 Generate N independent standard normals $Z_1, Z_2, \dots, Z_N \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$.
- 2 Set $B(0) = 0$.
- 3 For $i = 1, 2, \dots, N$: $B(i\Delta) = B((i-1)\Delta) + \sqrt{\Delta} Z_i$.

Each step adds an increment $\sqrt{\Delta} Z_i \sim \mathcal{N}(0, \Delta)$, which is exactly the correct distribution for the increment over a time step of length Δ .

5.2.1 Brownian Motion – Definition and Properties VI

Notation for Grid Simulation

Here Δ (Delta, meaning “step size”) $= T/N$ is the length of each time sub-interval; N is the total number of steps; Z_i (Z sub i) is the i -th independent standard normal draw; and $B(i\Delta)$ (B evaluated at time i times Delta) is the simulated BM value at the i -th grid point. In vector form, the path can be written as:

$$(B(\Delta), B(2\Delta), \dots, B(N\Delta)) \stackrel{d}{=} \sqrt{\Delta} (Z_1, Z_1 + Z_2, \dots, Z_1 + \dots + Z_N),$$

where $\stackrel{d}{=}$ (d with equals) means “has the same distribution as.”

5.2.1 Brownian Motion – Definition and Properties VII

Brownian Motion with Drift and Geometric Brownian Motion

BM with drift and diffusion coefficient:

$$X(t) = \mu t + \sigma B(t),$$

where μ (mu) is the drift (the average rate of growth per unit time) and σ (sigma) is the diffusion coefficient (measuring how strongly random fluctuations affect the path). This is used to model asset returns in a simple setting.

Geometric Brownian Motion under the risk-neutral pricing measure:

$$S(t) = S(0) \exp(\mu^* t + \sigma B(t)), \quad \mu^* = r - \frac{\sigma^2}{2}.$$

Here $S(t)$ (S of t) is the stock price at time t ; $S(0)$ is the price today; r (r) is the risk-free rate; μ^* (mu-star) is the *risk-neutral drift*, which is the risk-free rate adjusted downward by $\sigma^2/2$ to correct for the curvature (convexity) of the exponential function. With our parameters:

$$\mu^* = 0.05 - 0.0625/2 = -0.01875.$$

Why μ^* rather than μ ? The risk-neutral pricing principle says that under the “risk-neutral measure” (a technical probability measure used for derivative pricing), all assets grow at the risk-free rate r on average. The $\sigma^2/2$ correction arises from Itô's formula and can be seen as $\mathbb{E}[G(t)] = G(0) + rt$.

5.2.1 Brownian Motion – Python Code

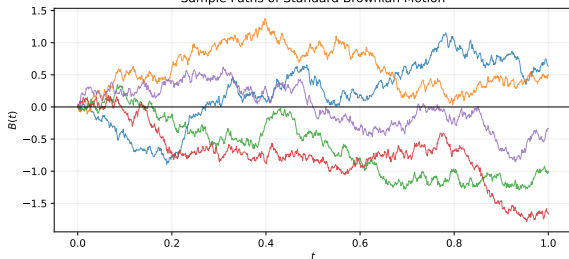
```
import numpy as np

def sim_brownian(N=1000, T=1.0, n_paths=5):
    """Simulate standard Brownian motion on [0,T] with N time steps."""
    dt = T / N
    t = np.linspace(0, T, N + 1)
    # Each row is one BM path; increments are sqrt(dt) * N(0,1)
    Z = np.random.standard_normal((n_paths, N))
    B = np.zeros((n_paths, N + 1))
    B[:, 1:] = np.cumsum(np.sqrt(dt) * Z, axis=1)
    return t, B

def sim_gbm(S0=100, r=0.05, sigma=0.25, T=1.0, N=252, n_paths=5):
    """Simulate Geometric Brownian Motion under risk-neutral measure.
    mu_star = r - sigma^2/2 is the risk-neutral drift.
    Each path: S(t_i) = S0 * exp(mu_star * t_i + sigma * B(t_i)).
    """
    dt = T / N
    mu_star = r - sigma**2 / 2 # risk-neutral drift
    Z = np.random.standard_normal((n_paths, N))
    # Cumulative log-return from time 0
    log_ret = np.cumsum(mu_star * dt + sigma * np.sqrt(dt) * Z, axis=1)
    log_S = np.log(S0) + np.concatenate(
        [np.zeros((n_paths, 1)), log_ret], axis=1)
    t = np.linspace(0, T, N + 1)
    return t, np.exp(log_S)
```

5.2.1 Brownian Motion Sample Paths

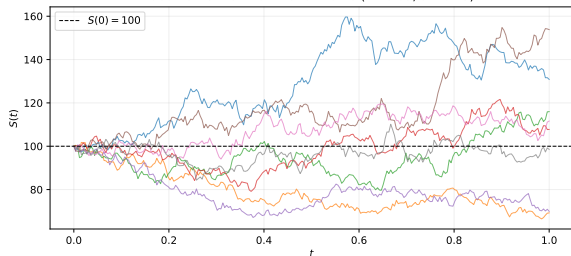
Sample Paths of Standard Brownian Motion



Standard BM $B(t)$: continuous paths, zero mean, variance = t growing linearly in time. No two paths ever look the same.

Notice that some GBM paths end well above $S(0) = 100$ while others end well below. The spread of outcomes is what makes option pricing interesting: if $S(T)$ were deterministic, no rational person would pay for the insurance an option provides.

Geometric Brownian Motion Paths ($r = 0.05$, $\sigma = 0.25$)



GBM $S(t) = S(0) \exp(\mu^*t + \sigma B(t))$: always positive; log-normally distributed at each t . The exponential ensures stock prices cannot go negative.

5.2.2 Financial Contracts and the Black-Scholes Formula I

A financial *option* is a contract that gives its holder the *right* (but not the obligation) to buy or sell an underlying asset at a specified price (the *strike*) on or before a specified date. The key question is: what is a fair price to pay for such a contract today?

Under the *risk-neutral pricing principle*, the fair price of any derivative equals its discounted expected payoff computed under the risk-neutral measure. In a GBM model, this means:

$$\text{Option price} = e^{-rT} \mathbb{E}[f(\text{stock path})],$$

where e^{-rT} (e to the minus r times T) is the *discount factor* (the present value of \$1 received at time T), and f (f) is the *payoff function* specific to the contract.

5.2.2 Financial Contracts and the Black-Scholes Formula II

Common Option Types Under GBM ($T = 1$ year)

- **European call:** Payoff $f = (S(1) - K)_+ = \max(S(1) - K, 0)$. The holder profits if the stock price at maturity $S(1)$ exceeds the strike K ; otherwise the option expires worthless.
- **European put:** Payoff $f = (K - S(1))_+$. Profits if the stock falls below the strike — a form of insurance.
- **Asian call:** Payoff $f = (A - K)_+$ where $A = \frac{1}{n} \sum_{j=1}^n S(j/n)$ is the *arithmetic average* stock price over n equally spaced monitoring dates. Asian options are cheaper than European calls and less susceptible to price manipulation at maturity.
- **Up-and-out call (barrier option):** Payoff $f = (S(1) - K)_+ \cdot \mathbf{1}(\max_{0 \leq t \leq 1} S(t) < b)$. This is a European call that is automatically cancelled (“knocked out”) if the stock price ever reaches the upper barrier b (b).
- **Basket call:** Payoff $f = (\sum_{j=1}^d w_j S_j(1) - K)_+$ for a portfolio of d stocks with weights w_j (w sub j).

5.2.2 Financial Contracts and the Black-Scholes Formula III

The Black-Scholes Formula (Theorem 5.2.7 – European Call)

For a European call with GBM stock price $S(t) = S(0) \exp(\mu^* t + \sigma B(t))$, the option price has the closed-form expression:

$$I_{\text{BS}} = S(0) \Phi(d_1) - K e^{-r} \Phi(d_2),$$

where Φ (Phi, capital phi) is the cumulative distribution function (CDF) of the standard normal distribution, i.e., $\Phi(x) = \mathbb{P}(Z \leq x)$ for $Z \sim \mathcal{N}(0, 1)$, and:

$$d_1 = \frac{1}{\sigma} \left[\ln \left(\frac{S(0)}{K} \right) + r + \frac{\sigma^2}{2} \right], \quad d_2 = d_1 - \sigma.$$

5.2.2 Financial Contracts and the Black-Scholes Formula IV

Notation for Black-Scholes

Here $\ln(S(0)/K)$ (natural logarithm of $S(0)$ over K) is the *log-moneyness* (positive if the stock is above the strike, negative if below); r (r) is the risk-free rate; σ (sigma) is the volatility; K (capital K) is the strike price; e^{-r} is the one-year discount factor. The quantity d_1 (d sub 1) can be thought of as a standardised measure of how far “in the money” the option is, after adjusting for time and volatility.

5.2.2 Financial Contracts and the Black-Scholes Formula V

Worked Numerical Example: Black-Scholes Price

With $S(0) = K = 100$ (at-the-money option), $r = 0.05$, $\sigma = 0.25$:

$$d_1 = \frac{1}{0.25} \left[\ln(100/100) + 0.05 + \frac{(0.25)^2}{2} \right] = \frac{1}{0.25} [0 + 0.05 + 0.03125] = \frac{0.08125}{0.25} = 0.325,$$

$$d_2 = 0.325 - 0.25 = 0.075.$$

Reading off the standard normal table (or computing numerically): $\Phi(0.325) \approx 0.6274$ and $\Phi(0.075) \approx 0.5299$. Therefore:

$$I_{BS} = 100 \times 0.6274 - 100 \times e^{-0.05} \times 0.5299 = 62.74 - 100 \times 0.9512 \times 0.5299 \approx 62.74 - 50.40 = 12.34.$$

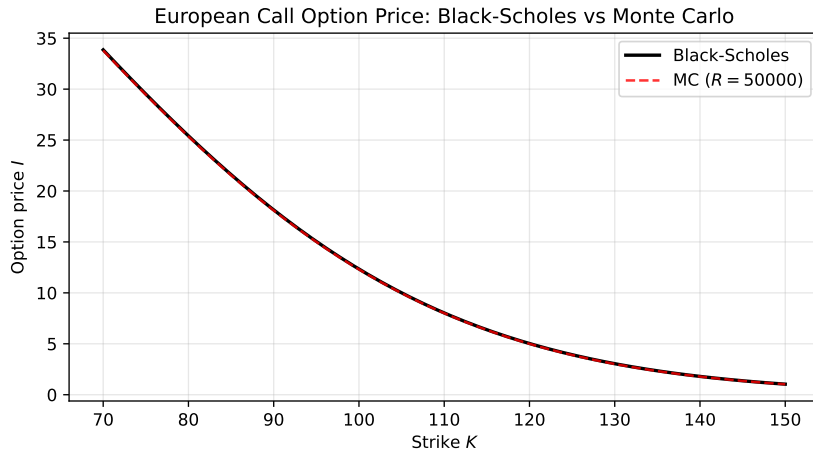
The exact European call price is approximately \$12.34. This is the benchmark against which all our Monte Carlo estimates will be compared.

5.2.2 Financial Contracts and the Black-Scholes Formula VI

Why Monte Carlo is Needed

The Black-Scholes formula applies *only* to the European call and put. For the Asian option, barrier option, and basket option, no comparable closed-form exists. Monte Carlo simulation is therefore the primary practical tool for pricing these contracts, which is why variance reduction matters so much in finance.

5.2.2 Black-Scholes Price vs. Strike



The Black-Scholes formula gives the exact European call price as a function of strike K (with all other parameters fixed). The curve is decreasing in K : a lower strike makes the option more valuable (easier to profit). When $K \gg S(0)$ (deep out-of-the-money, right side of the plot), the price is very small and CMC estimators waste most of their replications on zero payoff paths — making importance sampling and stratification essential.

5.2.3 Stratified Sampling for Option Pricing I

We now apply stratified sampling from Section 5.1.1 to option pricing. The key observation is that for the European call, the payoff depends on the stock price only through the final value $S(1)$, which in turn depends on $B(1)$ through the formula $S(1) = S(0) \exp(\mu^* + \sigma B(1))$. Since $B(1) \sim \mathcal{N}(0, 1)$, we can stratify on the standard normal quantile $Z = B(1)$.

5.2.3 Stratified Sampling for Option Pricing II

Example 5.2.5 – European Call with Stratified $Z = B(1)$

Divide the real line into m equally probable strata using the quantiles of $\mathcal{N}(0, 1)$:

$$A^j = \left(\Phi^{-1}\left(\frac{j-1}{m}\right), \Phi^{-1}\left(\frac{j}{m}\right) \right], \quad j = 1, 2, \dots, m,$$

where Φ^{-1} (Phi inverse) is the quantile function (inverse CDF) of the standard normal. Each stratum has probability $p_j = 1/m$.

Within stratum j : draw $U \sim \mathcal{U}[0, 1]$ and set $Z^j = \Phi^{-1}\left(\frac{j-1}{m} + \frac{U}{m}\right)$. This gives a sample Z^j that is uniformly spread within the j -th probability interval of the standard normal.

The stratified estimate of the call price:

$$\hat{I}_R^{\text{str}} = \frac{e^{-r}}{m} \sum_{j=1}^m \hat{Y}_{R_j}^j, \quad \hat{Y}_{R_j}^j = \frac{1}{R_j} \sum_{i=1}^{R_j} (S(0)e^{\mu^* + \sigma Z_i^j} - K)_+.$$

5.2.3 Stratified Sampling for Option Pricing III

Notation for Stratified Option Pricing

Here $\hat{Y}_{R_j}^j$ (Y-hat sub R-j superscript j) is the sample mean of the option payoff within stratum j ; R_j is the number of replications allocated to stratum j ; m is the total number of strata; the overall estimate $\hat{I}_R^{\text{str}}$ averages the stratum means with equal weight $1/m$ (since all strata have equal probability $p_j = 1/m$).

Numerical Results from Table 5.13 ($R = 10^5$, $S(0) = K = 100$, $r = 0.05$, $\sigma = 0.25$):

Strata m	CMC (no strata)	$m = 5$	$m = 50$	$m = 200$
Estimate $\hat{I}$	12.280	12.351	12.338	12.336
Half-width b	0.115	0.043	0.025	0.002
Variance reduction	$1\times$	$7.2\times$	$21\times$	$3\,778\times$

With $m = 200$ strata and optimal allocation, the variance is reduced by a factor of nearly 4 000. To achieve the same accuracy with CMC, we would need $4\,000 \times 10^5 = 4 \times 10^8$ replications instead of 10^5 — a dramatic saving. The key reason for this huge gain is that the European call payoff $(S(1) - K)_+$ is a monotone, smooth function of $Z = B(1)$, which stratified sampling handles very efficiently.

5.2.3 Stratified Sampling for Option Pricing IV

Example 5.2.8 – Asian Call with Stratified Brownian Motion

The Asian call payoff $f = e^{-r}(A - K)_+$ with $A = \frac{1}{n} \sum_{j=1}^n S(j/n)$ depends on the entire BM path, not just the endpoint. Stratifying a single input $Z = B(1)$ already helps:

Method 1 (single stratification): Stratify on $Z = B(1) \sim \mathcal{N}(0, 1)$. Given $Z = B(1) = z$, simulate the remaining path as a Brownian bridge conditioned on $B(1) = z$. This fixes the most influential source of randomness while leaving the bridge fluctuations free.

Method 2 (two-dimensional stratification): Use the bisection algorithm (Algorithm 38) to generate the path as $B(1), B(1/2), B(1/4), \dots$. Stratify jointly on the first two normal variates (Z_{2^k} and $Z_{2^{k-1}}$), creating m^2 two-dimensional strata.

Results: With $K = 100$ and $m = 200$ strata, variance is reduced by a factor of approximately $84\times$ for the Asian option. This is smaller than for the European option ($3\,778\times$) because the Asian payoff depends on multiple sources of randomness, not just $Z = B(1)$.

5.2.3 Stratified Sampling for Option Pricing V

Key Takeaway

Stratified sampling is most powerful when the payoff depends heavily on a small number of key random inputs. The European call depends only on $B(1)$, giving enormous gains. The Asian call depends on the full path, so gains are more modest but still substantial.

5.2.4 Control Variates for Asian Option Pricing I

The Asian call option uses the *arithmetic mean* $A = \frac{1}{n} \sum_{j=1}^n S(j/n)$ of the stock price. This has no known closed-form price. However, the closely related *geometric mean*

$$G = \left(\prod_{j=1}^n S(j/n) \right)^{1/n}$$

(G , the geometric mean — the n -th root of the product of all n price observations) *does* have a known closed-form price under GBM, because G is itself log-normally distributed when each $S(j/n)$ is log-normal.

The idea is to use the geometric-mean payoff $e^{-r}(G - K)_+$ as a control variate for the arithmetic-mean payoff $e^{-r}(A - K)_+$. Both depend on the same BM path, so they are highly correlated (large ρ), which means the control variate correction will be very effective.

5.2.4 Control Variates for Asian Option Pricing II

Control Variate: Geometric Mean (Example 5.2.9)

Under GBM, the geometric mean G satisfies:

$$\log G = \mu_N + \sigma_N Z, \quad Z \sim \mathcal{N}(0, 1),$$

where:

$$\mu_N = \mu^* \frac{n+1}{2n}, \quad \sigma_N^2 = \sigma^2 \frac{(n+1)(2n+1)}{6n^2}.$$

Here μ_N (μ sub N) is the effective drift of $\log G$, adjusted for the fact that G averages over n dates rather than using just the final price; σ_N^2 (σ squared sub N) is the effective variance of $\log G$. Since $G = S(0) \exp(\mu_N + \sigma_N Z)$ is log-normal, its discounted expected payoff $W = e^{-r} \mathbb{E}[(G - K)_+]$ can be computed via the Black-Scholes formula with parameters μ_N, σ_N :

$$W = e^{-r} [S(0) e^{\mu_N + \sigma_N^2/2} \Phi(d_1^G) - K \Phi(d_2^G)],$$

where d_1^G and d_2^G are the Black-Scholes d -values for the geometric Asian option. The control variate estimator for the arithmetic Asian option:

$$\hat{I}_R^{\text{cv}} = \hat{I}_R^{\text{CMC}} + \hat{c}(\hat{W}_R - W),$$

5.2.5 Antithetic Variates for the Asian Call Option I

We now apply the antithetic variates idea from Section 5.1.2 to the Asian call option. Recall the key principle: replace each normal random variable Z_j in the simulation by its negative $-Z_j$ to obtain a second, mirrored path. Since $-Z_j \sim \mathcal{N}(0, 1)$ whenever $Z_j \sim \mathcal{N}(0, 1)$, the mirrored path has the same statistical distribution as the original, but if one path tends upward, the mirrored path tends downward. When we average their payoffs, the random errors partially cancel.

5.2.5 Antithetic Variates for the Asian Call Option II

Construction of Antithetic Paths (Example 5.2.10)

Generate n independent standard normals $Z_1, Z_2, \dots, Z_n \sim \mathcal{N}(0, 1)$ (where n is the number of monitoring dates).

Original path (increment form):

$$B(k/n) = \frac{1}{\sqrt{n}} \sum_{j=1}^k Z_j, \quad k = 1, \dots, n.$$

Antithetic path (flip the sign of every Z_j):

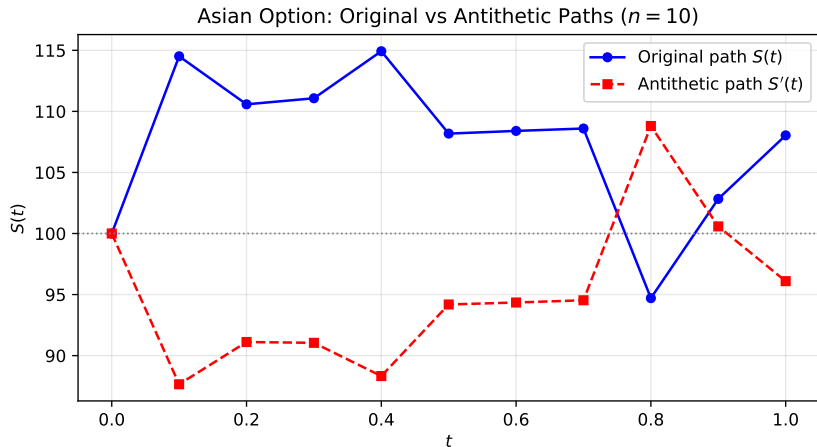
$$B'(k/n) = \frac{1}{\sqrt{n}} \sum_{j=1}^k (-Z_j) = -B(k/n).$$

Since $-Z_j \sim \mathcal{N}(0, 1)$ for all j , the antithetic path B' is a valid standard Brownian motion with the same distribution as B .

Define the original and antithetic GBM stock prices:

$$S(k/n) = S(0) \exp\left(\mu^*(k/n) + \sigma B(k/n)\right)$$

5.2.5 Antithetic Variates – Path Illustration



Each panel shows one pair of GBM paths: the original $S(t) = S(0) \exp(\mu^*t + \sigma B(t))$ and its antithetic mirror $S'(t) = S(0) \exp(\mu^*t - \sigma B(t))$. When one path trends upward, the other trends downward. The arithmetic means A and A' satisfy $\text{Corr}(A, A') < 0$, so their averaged payoff has much lower variance than either alone.

This illustrates the antithetic principle visually: errors in opposite directions cancel in the average.

5.2.6 Importance Sampling for the European Call I

Importance sampling shifts the sampling distribution towards the region where the option payoff is non-zero. For the European call, the payoff $(S(1) - K)_+$ is zero whenever the stock ends below the strike. If K is much larger than $S(0)$ (deep out-of-the-money), this happens most of the time under the standard normal distribution, making CMC very inefficient.

The fix is to draw $Z \sim \mathcal{N}(x^*, 1)$ (a normal centred at x^* instead of 0) so that samples are shifted toward the region where $S(1) > K$, and then reweight each sample by the likelihood ratio to correct for the shift.

5.2.6 Importance Sampling for the European Call II

IS Estimator for the European Call (Example 5.2.11)

The exact option price is:

$$I = e^{-r} \int_{-\infty}^{\infty} (S(0) e^{\mu^* + \sigma z} - K)_+ \phi_{0,1}(z) dz,$$

where $\phi_{0,1}(z)$ (ϕ sub 0-1 of z) is the standard normal probability density function evaluated at z . We instead draw $\tilde{Z} \sim \mathcal{N}(x^*, 1)$, i.e., from a normal centred at x^* (x -star).

The IS estimator is:

$$\hat{I}_R^{\text{IS}} = \frac{1}{R} \sum_{i=1}^R e^{-r} (S(0) e^{\mu^* + \sigma \tilde{Z}_i} - K)_+ \cdot \underbrace{e^{-x^* \tilde{Z}_i + (x^*)^2/2}}_{\text{likelihood ratio}},$$

where $\tilde{Z}_i \sim \mathcal{N}(x^*, 1)$ ($\tilde{Z}$ sub i) are independent samples from the shifted normal, and the term $e^{-x^* \tilde{Z}_i + (x^*)^2/2}$ is the likelihood ratio $\phi_{0,1}(\tilde{Z}_i)/\phi_{x^*,1}(\tilde{Z}_i)$ (the ratio of the target density to the proposal density at $\tilde{Z}_i$).

5.2.6 Importance Sampling for European Call – Python Code

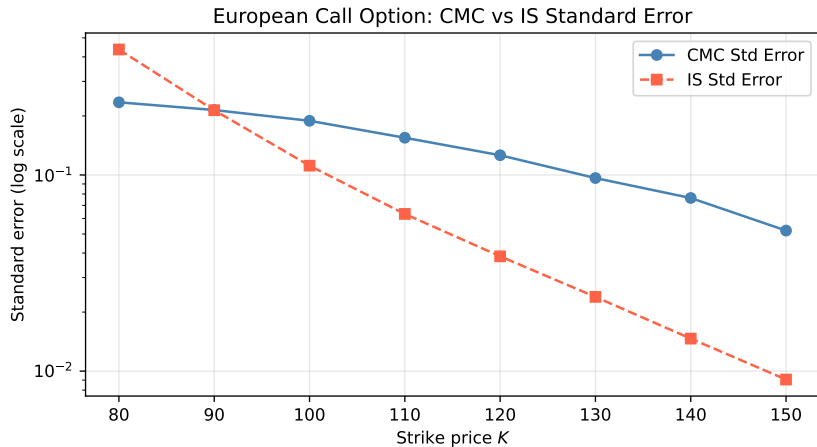
```
import numpy as np
from scipy.stats import norm

def european_call_IS_CE(S0=100, K=100, r=0.05, sigma=0.25,
                        R=10000, M=10000):
    """
    Estimate European call price via IS with CE-optimal shift x*.
    Step 1: pilot simulation to find x* = payoff-weighted mean of Z.
    Step 2: main IS simulation with proposal N(x*, 1).
    """
    mu_star = r - sigma**2 / 2  # risk-neutral drift (= -0.01875)

    # --- CE pilot: find the payoff-weighted mean of Z ---
    Z_pilot = np.random.standard_normal(M)
    S1_pilot = S0 * np.exp(mu_star + sigma * Z_pilot)
    payoff_pilot = np.maximum(S1_pilot - K, 0.0)
    # Weighted mean of Z using payoff as weight
    x_star = (np.sum(Z_pilot * payoff_pilot) /
              np.sum(payoff_pilot))
    print(f"    CE-optimal shift x* = {x_star:.4f}    (heuristic: ~1.033)")

    # --- Main IS simulation with proposal N(x_star, 1) ---
    Z_tilde = np.random.normal(x_star, 1.0, R)
    S1_IS = S0 * np.exp(mu_star + sigma * Z_tilde)
    payoff_IS = np.maximum(S1_IS - K, 0.0)
    # Log of likelihood ratio: log[phi(z;0,1) / phi(z;x*,1)]
    log_LR = -x_star * Z_tilde + x_star**2 / 2.0
```

5.2.6 IS for European Call – Standard Error vs. Strike



The plot shows the standard error of CMC vs. IS as a function of the strike K (with $S(0) = 100$ fixed). For at-the-money options ($K \approx 100$), CMC already performs reasonably and IS gives about 10 $\times$ improvement. For out-of-the-money options ($K > 110$), CMC's standard error remains large (many zero-payoff paths) while IS's error falls sharply, because the shift x^* moves the sampling distribution squarely into the payoff region. The

5.3 Bibliographical Notes I

This chapter draws on a rich body of literature in Monte Carlo methods, mathematical finance, and rare-event simulation. The following notes trace the key ideas to their original sources.

Variance reduction fundamentals and stratified sampling. The theoretical underpinnings of all the methods in Section 5.1 — including the law of total variance, stratification, antithetic variates, and the optimal coefficient for control variates — are developed in detail in Asmussen & Glynn *Stochastic Simulation: Algorithms and Analysis* [8] and in Rubinstein & Kroese *Simulation and the Monte Carlo Method* [186]. Both books contain many worked examples and proofs of the optimality results quoted in this chapter.

Antithetic variates, CRN, and monotonicity. The monotonicity condition for antithetic variates (Theorem 5.1.13 and Corollary 5.1.14) traces to early work by Hammersley and Morton in the 1950s. A modern treatment is in Asmussen & Glynn [8] and Ross *Simulation* [181]. The scheduling application of CRN (Example 5.1.20) is classical; the synchronisation requirement for CRN is discussed thoroughly in Asmussen & Glynn [8].

5.3 Bibliographical Notes II

Network reliability and rare-event importance sampling. The 11-node, 22-edge network reliability example is taken from Madras [138]. The logarithmic efficiency framework for rare-event IS (including the exponential change-of-measure and large deviations bounds) is developed in Asmussen & Glynn [8], Chapter VI.

Cross-entropy method. The CE method for IS parameter selection was introduced by Rubinstein [184] (1999) and is comprehensively treated in Rubinstein & Kroese [186]. An accessible tutorial is de Boer et al. [31], which includes many examples of applying CE to combinatorial optimisation and network reliability.

Brownian motion and simulation algorithms. The properties of BM (Definition 5.2.1 and Proposition 5.2.2) are classical results found in any graduate probability text; see, e.g., Durrett *Probability: Theory and Examples*. The bisection (Brownian bridge) simulation algorithm is discussed in Asmussen & Glynn [8] and Knuth [106].

Mathematical finance and the Black-Scholes model. The Black-Scholes formula (Theorem 5.2.7) is from Black & Scholes (1973) and is derived in many standard texts including Hull *Options, Futures, and Other Derivatives* [96]. The risk-neutral pricing framework used here is standard; see also Shreve *Stochastic Calculus for Finance* (2004).

5.3 Bibliographical Notes III

Asian option pricing. The geometric mean as a control variate for the arithmetic Asian option was proposed by Kemna & Vorst (1990); the derivation of the geometric mean's distribution under GBM follows directly from the log-normality of each price observation. A comprehensive treatment of variance reduction for Asian options — including all four methods covered in Section 5.2 — is in Glasserman *Monte Carlo Methods in Financial Engineering* [74], Chapter 4.

5.4 Selected Exercises I

The exercises below range from analytical proofs to numerical implementations. Working through them in order builds understanding from the theoretical foundations (exercises 5.T.1 and 5.T.5) through the practical optimality conditions (5.T.8 and 5.T.25) to hands-on computation (Lab 5.L.1).

Exercise 5.T.1 – Stratified Sampling with Unequal Costs

In practice, different strata may require different computation times. Suppose we estimate $I = \mathbb{E}[Y]$ using strata $A^1, \dots, A^m$ with stratum probabilities $p_j = \mathbb{P}(Y \in A^j) > 0$ and within-stratum standard deviations $\sigma_j = \sqrt{\text{Var}(Y \mid Y \in A^j)}$. Suppose additionally that the cost (computation time) per replication in stratum j is $t_j > 0$ (t sub j), and our total budget is C (capital C, total available CPU time, not total number of replications).

- Let $\gamma_j = \lim_{R \rightarrow \infty} R_j t_j / C$ be the fraction of the budget allocated to stratum j (gamma sub j), so $\sum_j \gamma_j = 1$. Show that the stratified estimator satisfies a CLT with asymptotic variance:

$$\zeta^2 = \left(\sum_{j=1}^m \frac{p_j^2 \sigma_j^2}{\gamma_j} \right) \left(\sum_{k=1}^m t_k \gamma_k \right).$$

5.4 Selected Exercises II

Here ζ^2 (zeta squared) is the time-normalised asymptotic variance (variance times total computation time).

- ⓑ Show that the budget allocation minimising ζ^2 subject to $\sum_j \gamma_j = 1$ and $\gamma_j \geq 0$ is:

$$\gamma_j^* = \frac{p_j \sigma_j / \sqrt{t_j}}{\sum_{k=1}^m p_k \sigma_k / \sqrt{t_k}}.$$

Interpret this result: more budget goes to strata that are more variable (σ_j large) *and* cheaper to simulate (t_j small). Explain why both factors make intuitive sense.

5.4 Selected Exercises III

Exercise 5.T.5 – Antithetic Variates for the Exponential Function

Let $Y = e^U$ where $U \sim \mathcal{U}[0, 1]$. The true value is $I = \mathbb{E}[e^U] = e - 1 \approx 1.7183$.

- (a) Compute $\text{Var}(e^U) = \mathbb{E}[e^{2U}] - (\mathbb{E}[e^U])^2$. Use $\mathbb{E}[e^{2U}] = \int_0^1 e^{2u} du = (e^2 - 1)/2$. *Answer:* $\text{Var}(e^U) = (e^2 - 1)/2 - (e - 1)^2 \approx 0.2420$.
- (b) Compute $\text{Cov}(e^U, e^{1-U})$. Use the hint: $\mathbb{E}[e^U \cdot e^{1-U}] = \mathbb{E}[e^1] = e$ (since $U + (1 - U) = 1$ always). Therefore: $\text{Cov}(e^U, e^{1-U}) = e - (e - 1)^2 = e - e^2 + 2e - 1 = 3e - 1 - e^2$. Numerically: $\approx 3(2.718) - 1 - 7.389 \approx -0.2342$.
- (c) Compute the variance of the antithetic average $(e^U + e^{1-U})/2$:

$$\text{Var}\left(\frac{e^U + e^{1-U}}{2}\right) = \frac{\text{Var}(e^U) + \text{Cov}(e^U, e^{1-U})}{2} = \frac{0.2420 + (-0.2342)}{2} \approx 0.00390.$$

Confirm that the variance reduction factor is $0.2420/(2 \times 0.00390) \approx 31\times$.

- (d) Compute $\text{Corr}(e^U, e^{1-U}) = \text{Cov}/\text{Var}$ and verify that $1 + \text{Corr} \approx 0.032$, consistent with a $\approx 31\times$ variance reduction.

5.4 Selected Exercises IV

Exercise 5.T.8 – Optimal IS Proposal is the Zero-Variance Density

Let $k : E \rightarrow \mathbb{R}_{\geq 0}$ be a non-negative function and let f be the target density. Define the *zero-variance IS proposal*:

$$g^*(x) = \frac{k(x)f(x)}{\int_E k(x)f(x) dx} = \frac{k(x)f(x)}{I}.$$

Show that g^* achieves zero variance for the IS estimator, in the sense that $\text{Var}_{g^*}[k(\tilde{X})f(\tilde{X})/g^*(\tilde{X})] = 0$.

Proof sketch: Under proposal g^* , the IS weight for each sample $\tilde{X} \sim g^*$ is:

$$k(\tilde{X}) \frac{f(\tilde{X})}{g^*(\tilde{X})} = k(\tilde{X}) \frac{f(\tilde{X})}{k(\tilde{X})f(\tilde{X})/I} = I,$$

which is the constant I regardless of $\tilde{X}$. A constant random variable has variance exactly zero, so the IS estimator based on g^* returns the exact value I from a single sample. (The catch: to use g^* we must know I , which is what we want to compute — a circular dependency.)

5.4 Selected Exercises V

Then use Jensen's inequality to show that for any other proposal g :

$$\mathbb{E}_g \left[\left(\frac{k(\tilde{X})f(\tilde{X})}{g(\tilde{X})} \right)^2 \right] \geq \left(\mathbb{E}_g \left[\frac{k(\tilde{X})f(\tilde{X})}{g(\tilde{X})} \right] \right)^2 = I^2,$$

with equality if and only if $k(x)f(x)/g(x) = \text{const}$ for $\mathbb{P}_g$ -almost all x , i.e., iff $g = g^*$.

5.4 Selected Exercises VI

Exercise 5.T.25 – CE-Optimal Parameter for the Normal Family

For $I = \mathbb{P}(Z > \gamma)$ where $Z \sim \mathcal{N}(0, 1)$, consider the IS family $\{\mathcal{N}(\theta, 1) : \theta \in \mathbb{R}\}$. The CE objective is to maximise over θ (theta):

$$D(\theta) = \mathbb{E}_{\mathcal{N}(0,1)}[\mathbf{1}(Z > \gamma) \log \phi_{\theta,1}(Z)],$$

where $\phi_{\theta,1}(z) = \frac{1}{\sqrt{2\pi}} \exp(-(z - \theta)^2/2)$ is the $\mathcal{N}(\theta, 1)$ density.

- Ⓐ Expand $\log \phi_{\theta,1}(z) = -\frac{1}{2} \log(2\pi) - \frac{(z-\theta)^2}{2}$ and show that maximising $D(\theta)$ over θ is equivalent to maximising $\theta \mathbb{E}[\mathbf{1}(Z > \gamma) Z]$.
- Ⓑ Conclude that the CE-optimal parameter is:

$$\theta^* = \frac{\mathbb{E}[\mathbf{1}(Z > \gamma) Z]}{\mathbb{P}(Z > \gamma)} = \mathbb{E}[Z \mid Z > \gamma],$$

the mean of $\mathcal{N}(0, 1)$ truncated to (γ, ∞) .

5.4 Selected Exercises VII

- Ⓢ Using the formula $\mathbb{E}[Z \mid Z > \gamma] = \phi(\gamma)/(1 - \Phi(\gamma))$ where ϕ (phi, lowercase) is the standard normal density and Φ (Phi, uppercase) is the CDF, show:

$$\theta^* = \frac{e^{-\gamma^2/2}/\sqrt{2\pi}}{1 - \Phi(\gamma)} = \frac{\sqrt{2} e^{-\gamma^2/2}}{\sqrt{\pi}(1 - \operatorname{erf}(\gamma/\sqrt{2}))},$$

where $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$ (the error function). Verify numerically that for $\gamma = 4$: $\theta^* = 4.2256$.

5.4 Selected Exercises VIII

Lab Exercise 5.L.1 – Antithetic Variates for $\int_0^1 e^x dx$

Implement and compare the following two estimators for $I = \int_0^1 e^x dx = e - 1$ using Python. Use $R = 10\,000$ replications (and repeat each estimator 2 000 times to estimate empirical variances):

- ➊ **CMC:** Generate $U_1, \dots, U_R \sim \mathcal{U}[0, 1]$, return $\hat{I}^{\text{CMC}} = \frac{1}{R} \sum_{i=1}^R e^{U_i}$.
- ➋ **Antithetic:** Generate $U_1, \dots, U_{R/2} \sim \mathcal{U}[0, 1]$, return $\hat{I}^{\text{anti}} = \frac{1}{R/2} \sum_{i=1}^{R/2} \frac{e^{U_i} + e^{1-U_i}}{2}$.

Tasks:

- ➐ Report the empirical standard deviation of $\hat{I}^{\text{CMC}}$ and $\hat{I}^{\text{anti}}$ over the 2 000 repetitions.
- ➑ Compute the empirical variance reduction factor and compare it to the theoretical value derived in Exercise 5.T.5.
- ➒ Plot a histogram of the 2 000 estimates from each method on the same axes. The antithetic histogram should be much narrower.
- ➓ Repeat with $R = 1\,000$ and $R = 100\,000$ and comment on how the variance reduction factor changes (it should remain approximately constant).

Chapter 5 Summary I

Chapter 5 Summary II

Core Message

Every variance reduction technique in this chapter rests on the same fundamental idea: find a way to extract more useful information from each simulation run, so that the estimator's variance is smaller than that of the crude Monte Carlo average. The methods differ in how they achieve this.

- **Stratified sampling** guarantees that every region of the input space is sampled, eliminating the “luck of the draw” in how samples are distributed across regions. It is most powerful for smooth, monotone integrands.
- **Antithetic variates** generates pairs of negatively correlated estimates so that their average has lower variance than either alone. It works when the integrand is monotone in the input uniform variables.
- **Common random numbers** induces positive correlation between two system outputs so that their difference (which is what we want to estimate) has lower variance. It is the tool of choice when comparing two designs.
- **Control variates** corrects the crude estimate using the observed deviation of a correlated auxiliary variable from its known mean. The stronger the correlation, the greater the reduction.
- **Conditional Monte Carlo (Rao-Blackwell)** replaces noisy single observations by their analytical conditional means, always reducing variance.